Q1: The relation between time t and distance x for a moving body is given as $t = mx^2 + nx$, where m and n are constants. The retardation of the motion is : (When v stands for velocity)

(A) $2\,mv^3$

(B) $2\,mnv^3$

(C) $2nv^3$

(D) $2n^2v^3$

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(D) $2n^2v^3$

Solution:

$$t = mx^2 + nx$$
$$\Rightarrow 1 = m \times 2xv + n \times v$$
$$\Rightarrow v = \frac{1}{2mx+n}$$
$$\therefore \frac{dv}{dt} = -\frac{1}{(2mx+n)^2} \times (2m) \times v$$
$$\Rightarrow a = -(2m) \times v \times v^2$$
$$= -2mv^3$$

Q2: In a simple harmonic oscillation, what fraction of total mechanical energy is in the form of kinetic energy, when the particle is midway between mean and extreme position.

(A) $\frac{1}{2}$

(B) $\frac{3}{4}$

(C) $\frac{1}{3}$

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(D) $\frac{1}{4}$

Solution:

$$U = \frac{1}{2} m\omega^2 A^2$$

$$v = \omega\sqrt{A^2 - \frac{A^2}{4}}$$

$$k = \frac{1}{2} mv^2 = \frac{1}{2} m\omega^2 \times \frac{3A^2}{4}$$

Q3: A force $\vec{F} = \left(40\hat{i} + 10\hat{j}\right) N$ acts on a body of mass 5 kg. If the body starts from rest, its position vector $\vec{r}\ at\ time\ t = 10s$, will be

(A) $\left(100\hat{i} + 400\hat{j}\right) m$

(B) $\left(100\hat{i} + 100\hat{j}\right) m$

(C) $\left(400\hat{i} + 100\hat{j}\right) m$

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(D) $\left(400\hat{i} + 400\hat{j}\right) m$

Solution:

$$\vec{r} = \vec{u}t + \frac{1}{2}\vec{a}t^2$$

$$= \frac{1}{2} \times \left(8\hat{i} + 2\hat{j}\right) \times 100$$

$$= \left(400\hat{i} + 100\hat{j}\right) m$$

Q4: A prism of refractive index μ and angle of prism A is placed in the position of minimum angle of deviation. If minimum angle of deviation is also A, then in terms of refractive index

(A) $2\cos^{-1}\left(\frac{\mu}{2}\right)$

(B) $\sin^{-1}\left(\frac{\mu}{2}\right)$

(C) $\sin^{-1}\left(\sqrt{\frac{\mu-1}{2}}\right)$

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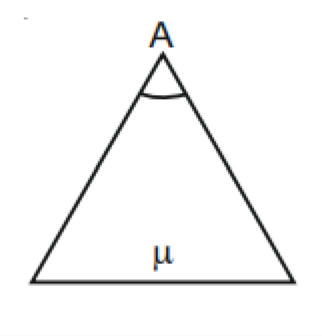
Solution:



$$\delta_m = A$$

$$\therefore \mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

$$\Rightarrow \mu = \frac{\sin A}{\sin\left(\frac{A}{2}\right)} = 2\cos\left(\frac{A}{2}\right)$$

$$\Rightarrow \frac{A}{2} = \cos^{-1}\left(\frac{\mu}{2}\right)$$

$$\Rightarrow A = 2\cos^{-1}\left(\frac{\mu}{2}\right)$$

Q5: A heat engine has an efficiency of $\frac{1}{6}$. When the temperature of sink is reduced by 62°C, its efficiency get doubled. The temperature of the source is :

(A) $124^o C$

(B) $37^o C$

(C) $62^o C$

(D) $99^o C$

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Solution:

$\eta = 1 - \frac{T_L}{T_H}$

$\Rightarrow \frac{1}{6} = 1 - \frac{T_L}{T_H} \Rightarrow \frac{T_L}{T_H} = \frac{5}{6}$

Now, $2 \times \frac{1}{6} = 1 - \frac{\mathrm{T_L - 62}}{\mathrm{T_H}} \Rightarrow \frac{\mathrm{T_L - 62}}{\mathrm{T_H}} = \frac{2}{3}$ ...(ii) from (i) and (ii)

$\frac{T_L}{T_L - 62} = \frac{5}{6} \times \frac{3}{2} = \frac{5}{4}$

$4T_L = 5T_L - 310$

$T_L = 310 \mathrm{~K}$

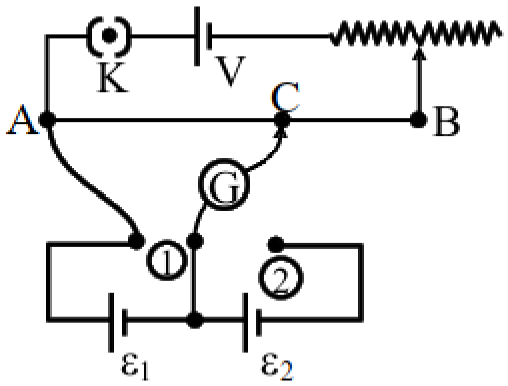
from (i)

$T_H = \frac{6}{5} \times T_L = \frac{6}{5} \times 310 = 372 \mathrm{~K}$

$= (372 - 273)^\circ \mathrm{C}$

$= 99^\circ \mathrm{C}$

Q6: In the given potentiometer circuit arrangement, the balancing length AC is measured to be 250 cm. When the galvanometer connection is shifted from point (1) to point (2) in the given diagram, the balancing length becomes 400 cm. The ratio of the emf of two cells, $\frac{\varepsilon_1}{\varepsilon_2}$ is :



(A) $\frac{5}{3}$

(B) $\frac{8}{5}$

(C) $\frac{4}{3}$

(D) $\frac{3}{2}$

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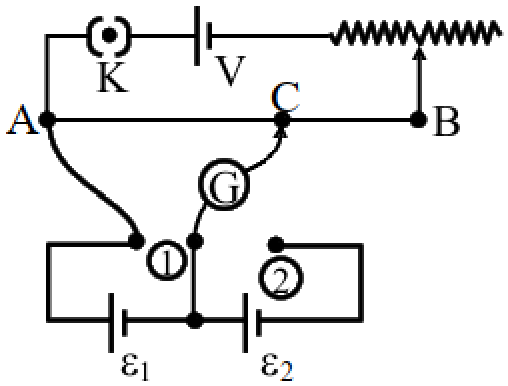

(A) $\frac{5}{3}$

(B) $\frac{8}{5}$

(C) $\frac{4}{3}$

(D) $\frac{3}{2}$

Solution:

$$\varepsilon_1 \propto 250$$
$$\varepsilon_1 + \varepsilon_2 \propto 400$$
$$\Rightarrow \frac{\varepsilon_1+\varepsilon_2}{\varepsilon_1} = \frac{400}{250} = \frac{8}{5}$$
$$\Rightarrow 1 + \frac{\varepsilon_2}{\varepsilon_1} = \frac{8}{5}$$
$$\Rightarrow \frac{\varepsilon_2}{\varepsilon_1} = \frac{3}{5} \Rightarrow \frac{\varepsilon_1}{\varepsilon_2} = \frac{5}{3}$$

Q7: Two ions having same mass have charges in the ratio $1 : 2$. They are projected normally in a uniform magnetic field with their speeds in the ratio $2 : 3$. The ratio of the radii of their circular trajectories is:

(A) $1 : 4$

(B) $4 : 3$

(C) $3 : 1$

(D) $2 : 3$

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(A) $1:4$

(B) $4:3$

(C) $3:1$

(D) $2:3$

Solution:

$$\mathrm{R}=\frac{mv}{q\,\mathrm{B}} \quad \frac{q_1}{q_2}=\frac{1}{2}, \frac{v_1}{v_2}=\frac{2}{3}$$
$$\frac{\mathrm{R}_1}{\mathrm{R}_2}=\frac{v_1}{q_1}\times\frac{q_2}{v_2}=\frac{v_1}{v_2}\times\frac{q_2}{q_1}$$
$$=\frac{2}{3}\times 2=\frac{4}{3}$$

Q8: A 10Ω resistance is connected across $220V-50Hz$AC supply. The time taken by the current to change from its maximum value to the rms value is:

(A) $2.5 \mathrm{~ms}$

(B) $1.5 \mathrm{~ms}$

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(D) $4.5 \mathrm{~ms}$

Solution:

$i=\frac{220}{\mathrm{R}} \sin \omega t, \mathrm{~T}=\frac{1}{50} \mathrm{~s}$

time required for maximum to rms

$=\frac{\mathrm{T}}{8}$

$=\frac{1}{50 \times 8} \mathrm{~s}$

$=\frac{1}{400} \mathrm{~s}$

$=2.5 \mathrm{~ms}$

Q9: A balloon was moving upwards with a uniform velocity of 10 m/s. An object of finite mass is dropped from the balloon when it was at a height of 75 m from the ground level. The height of the balloon from the ground when object strikes the ground was around :
(takes the value of g as $10\ m/s^2$)

(A) $300\ \mathrm{m}$

(B) $200\ \mathrm{m}$

(C) $125\ \mathrm{m}$

(D) $250\ \mathrm{m}$

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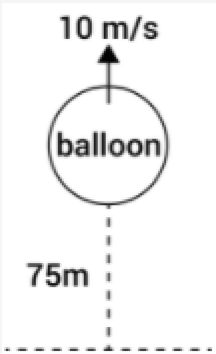

$$-75 = 10t - \frac{1}{2} \times (10)t^2$$

$$\Rightarrow -15 = 2t - t^2$$

$$\Rightarrow t^2 - 2t - 15 = 0$$

$$\Rightarrow t = 5 \text{ s}$$

$$\mathrm{H} = 75 + 10 \times 5 = 125 \text{ m}$$

Q10: If q_f is the free charge on the capacitor plates and q_b is the bound charge on the dielectric slab of dielectric constant k placed between the capacitor plates, then bound charge q_b can be expressed as :

(A) $q_b = q_f\left(1 - \frac{1}{\sqrt{k}}\right)$

(B) $q_b = q_f\left(1 - \frac{1}{k}\right)$

(C) $q_b = q_f\left(1 + \frac{1}{\sqrt{k}}\right)$

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Solution:

We know
$q_b = q_f\left(1 - \frac{1}{k}\right)$

Q11: Consider a planet in some solar system which has a mass double the mass of earth and density equal to the average density of earth. If the weight of an object on earth is W, the weight of the same object on that planet will be :

(A) 2W

(B) W

(C) $2^{\frac{1}{3}}W$

(D) $\sqrt{2}W$

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(D) $\sqrt{2} W$

Solution:

$$g_{\text{eff}} = \frac{\mathrm{GM}}{\mathrm{R}^2}$$

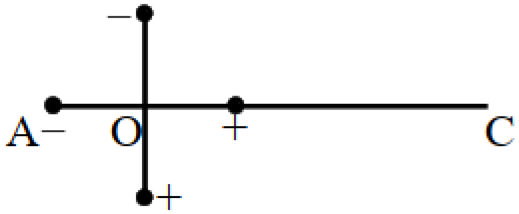
$$2 \times \rho \times \frac{4}{3} \pi \mathrm{R}_1^3 = \frac{4}{3} \pi \mathrm{R}_2^3 \times \rho$$

$$\mathrm{R}_2 = 2^{1/3} R_1$$

$$g_1 = \frac{\mathrm{GM}}{{R_1}^2}$$

$$g_2 = \frac{2\mathrm{GM}}{\left[2^{1/3} R_1\right]^2} = \frac{2^{1/3}\mathrm{GM}}{{R_1}^2}$$

Q12: Two ideal electric dipoles A and B, having their dipole moment p_1 and p_2 respectively are placed on a plane with their centres at O as shown in the figure. At point C on the axis of dipole A, the resultant electric field is making an angle of 37° with the axis. The ratio of the dipole moment of A and B, $\frac{p_1}{p_2}$ is : $(take\ sin\ 37^o = \frac{3}{5})$



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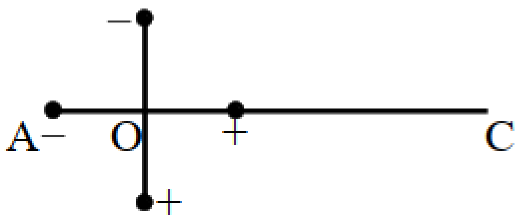

(A) $\frac{3}{8}$

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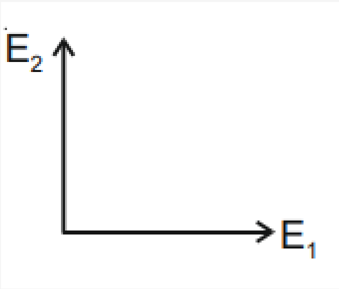
(D) $\frac{4}{3}$

Solution:



$$\overrightarrow{\mathrm{E}}_{\mathrm{C}} = \overrightarrow{\mathrm{E}}_{1} + \overrightarrow{\mathrm{E}}_{2}$$

$$\mathrm{E}_1 = \frac{2p_1}{4\pi\varepsilon_0 r^3}\hat{i}$$

$$\mathrm{E}_2 = \frac{p_2}{4\pi\varepsilon_0 r^3}\hat{j}$$

$$\tan 37^\circ = \frac{E_2}{E_1} = \frac{p_2}{2p_1}$$

$$\frac{3}{4} = \frac{p_2}{2p_1}$$

$$\frac{p_1}{p_2} = \frac{2}{3}$$

Q13: Two spherical soap bubbles of radii r_1 and r_2 in vacuum combine under isothermal conditions. The resulting bubble has a radius equal to :

(A) $\frac{r_1 r_2}{r_1+r_2}$

(B) $\sqrt{r_1 r_2}$

(C) $\sqrt{r_1^2+r_2^2}$

(D) $\frac{r_1+r_2}{2}$

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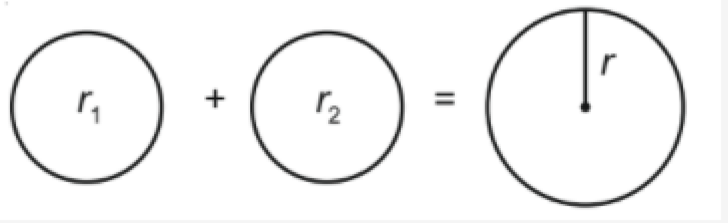
Solution:



$$P_1 = P_0 + \frac{4T}{r_1}, P_2 = P_0 + \frac{4T}{r_2}, P = P_0 + \frac{4T}{r}$$

$$and\, n_1 + n_2 = n$$

$$\Rightarrow \frac{P_1 V_1}{RT} + \frac{P_2 V_2}{RT} = \frac{P \times V}{RT}$$

$$\Rightarrow r_1^2 + r_2^2 = r^2$$

$$\Rightarrow r = \sqrt{r_1^2 + r_2^2}$$

Q14: The force is given in terms of time t and displacement x by the equation $F = A\ cos\ Bx + C\ sin\ Dt$ The dimensional formula of $\frac{AD}{B}$ is :

(A) $\left[M^0 LT^{-1}\right]$

(B) $\left[ML^2 T^{-3}\right]$

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(D) $\left[M^2 L^2 T^{-3}\right]$

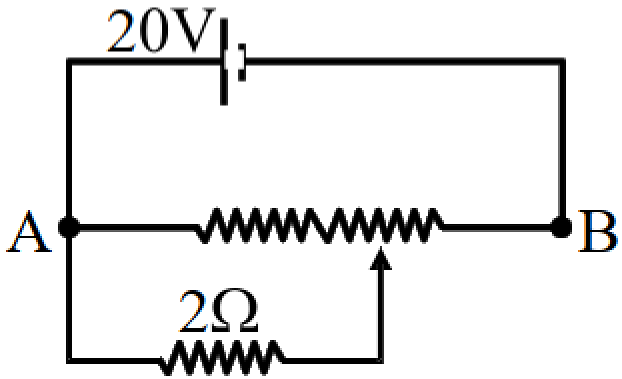
Solution:

$$F = A\cos Bx + C\sin Dt$$

$$\therefore \left[\frac{AD}{B}\right] = \frac{MLT^{-2} \times T^{-1}}{L^{-1}}$$

$$= \left[\mathrm{ML}^2 T^{-3}\right]$$

Q15: The given potentiometer has its wire of resistance 10Ω. When the sliding contact is in the middle of the potentiometer wire, the potential drop across 2Ω resistor is :



(A) $10V$

(B) $5V$

(C) $\frac{40}{9}V$

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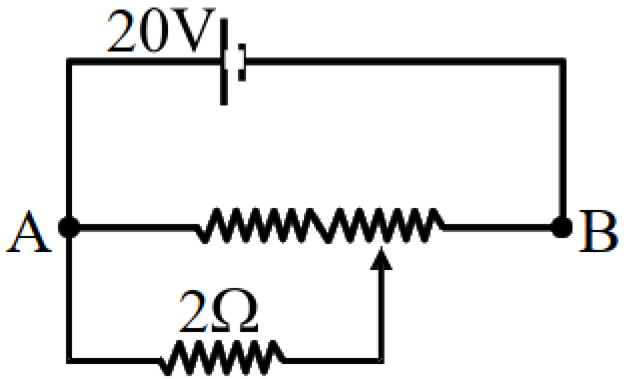

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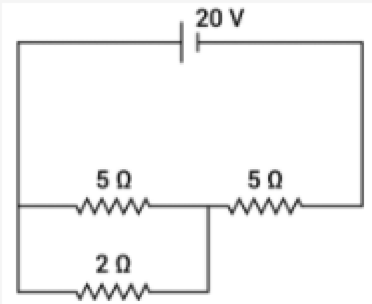

$$R_1 = \frac{10}{7}, R_2 = 5$$

$$\Rightarrow \frac{R_1}{R_2} = \frac{2}{7},\ V_{2\Omega} = \frac{2}{9} \times 20 = \frac{40}{9}\ \text{V}$$

Q16: An electron moving with speed v and a photon moving with speed c, have same D-Broglie wavelength. The ratio of kinetic energy of electron to that of photon is :

(A) $\frac{3c}{v}$

(B) $\frac{v}{3c}$

(C) $\frac{v}{2c}$

(D) $\frac{2c}{v}$

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(B) $\frac{v}{3c}$

(C) $\frac{v}{2c}$

(D) $\frac{2c}{v}$

Solution:

$$\frac{h}{mv} = \lambda$$

$$k_e = \frac{h^2}{2m\lambda^2}$$

$$k_{ph} = \frac{hc}{\lambda}$$

$$\frac{k_e}{k_{ph}} = \frac{h^2}{2m\lambda^2} \times \frac{\lambda}{hc}$$

$$\text{ratio} = \frac{h}{2m\lambda c} = \frac{h}{2mc \times \frac{h}{mv}}$$

$$= \frac{v}{2c}$$

Q17: The instantaneous velocity of a particle moving in a straight line is given as $v = \alpha t + \beta t^2$, where α and β are constants. The distance travelled by the particle between 1s and 2s is :

(A) $3\alpha + 7\beta$

(B) $\frac{3}{2}\alpha + \frac{7}{3}\beta$

(C) $\frac{\alpha}{2} + \frac{\beta}{3}$

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(D) $\frac{3}{2}\alpha + \frac{7}{2}\beta$

Solution:

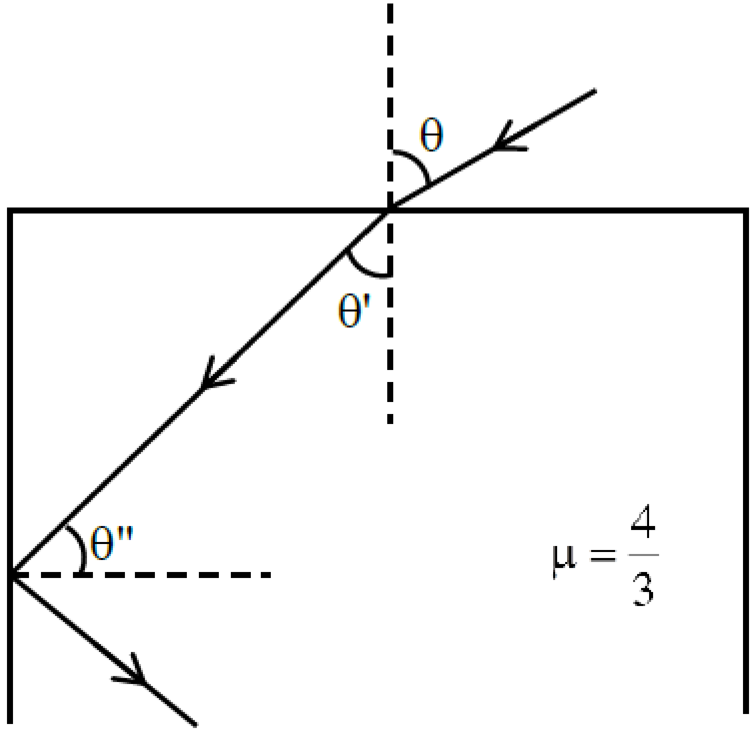
$$v = \alpha t + \beta t^2$$

$$\int dx = \int_1^2 \left(\alpha t + \beta t^2\right) dt$$

$$= \alpha \frac{t^2}{2}\Big]_1^2 + \frac{\beta}{3} t^3\Big]_1^2$$

$$= \frac{3}{2}\alpha + \frac{7}{3}\beta$$

Q18: A ray of light entering from air into a denser medium of refractive index $\frac{4}{3}$, as shown in figure. The light ray suffers total internal reflection at the adjacent surface as shown. The maximum value of angle θshould be equal to :



(A) $\sin^{-1}\frac{\sqrt{7}}{3}$

(B) $\sin^{-1}\frac{\sqrt{5}}{4}$

(C) $\sin^{-1}\frac{\sqrt{7}}{4}$

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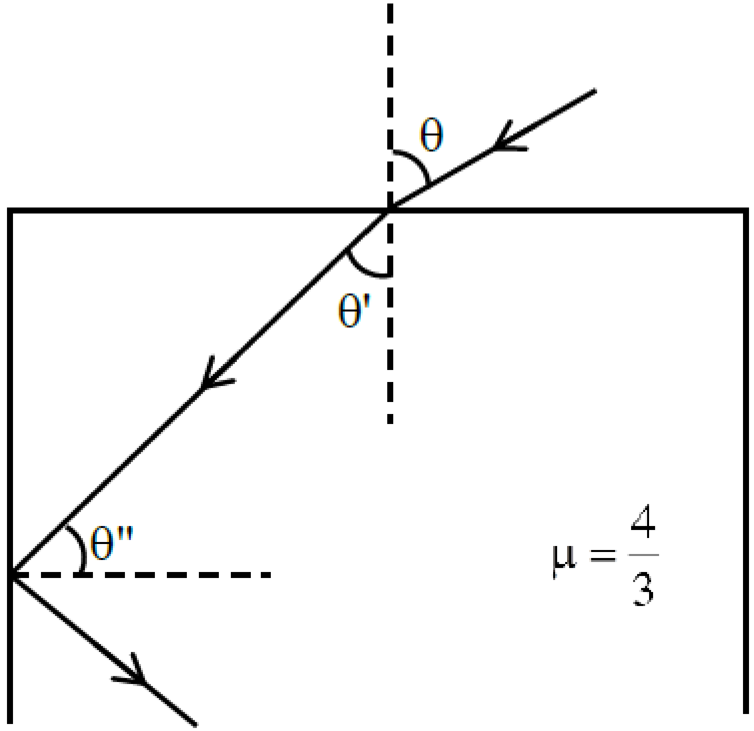

(A) $\sin^{-1}\frac{\sqrt{7}}{3}$

(B) $\sin^{-1}\frac{\sqrt{5}}{4}$

(C) $\sin^{-1}\frac{\sqrt{7}}{4}$

(D) $\sin^{-1}\frac{\sqrt{5}}{3}$

Solution:

$$\frac{4}{3}\sin\theta'' = 1$$

$$\Rightarrow \sin\theta'' = \frac{3}{4}$$

$$\frac{4}{3}\sin\theta' = 1 \times \sin\theta$$

$$\Rightarrow \sin\theta = \frac{4}{3}\sqrt{1-\frac{9}{16}}$$

$$= \frac{4}{3}\sqrt{\frac{7}{16}} = \frac{\sqrt{7}}{3}$$

Q19: When radiation of wavelength λ is incident on a metallic surface, the stopping potential of ejected photoelectrons is 4.8 V. If the same surface is illuminated by radiation of double the previous wavelength, then the stopping potential becomes 1.6 V. The threshold wavelength of the metal is :

(A) 2λ

(B) 4λ

(C) 8λ

(D) 6λ

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(A) 2λ

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Solution:

$$\frac{\mathrm{hc}}{\lambda}=\frac{\mathrm{hc}}{\lambda_0}+4.8\ldots(i)$$

$$\frac{\mathrm{hc}}{2\lambda}=\frac{\mathrm{hc}}{\lambda_0}+1.6\ldots(ii)$$

Eq. (i) ÷eq. (ii),

$$\Rightarrow 2=\frac{\frac{hc}{\lambda_0}+4.8}{\frac{hc}{\lambda_0}+1.6}$$

$$\Rightarrow \frac{hc}{\lambda_0}=1.6 \Rightarrow \lambda_0=4\lambda$$

Q20: Two vectors $\vec{X}$and $\vec{Y}$have equal magnitude. The magnitude of $\left(\vec{X}-\vec{Y}\right)$ is n times the magnitude of $\left(\vec{X}+\vec{Y}\right)$. The angle between $\vec{X}$and $\vec{Y}$is :

(A) $\cos^{-1}\left(\frac{-n^2-1}{n^2-1}\right)$

(B) $\cos^{-1}\left(\frac{n^2-1}{-n^2-1}\right)$

(C) $\cos^{-1}\left(\frac{n^2+1}{-n^2-1}\right)$

(D) $\cos^{-1}\left(\frac{n^2+1}{n^2-1}\right)$

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(D) $\cos^{-1}\left(\frac{n^2+1}{n^2-1}\right)$

Solution:

$$|\vec{X}+\vec{Y}|^2 = X^2+Y^2+2XY\cos\theta$$
$$|\vec{X}-\vec{Y}|^2 = X^2+Y^2-2XY\cos\theta$$
$$n^2 = \frac{2-2\cos\theta}{2+2\cos\theta}$$
$$n^2 = \frac{1-\cos\theta}{1+\cos\theta}$$
$$n^2+n^2\cos\theta = 1-\cos\theta$$
$$\cos\theta = -\left(\frac{n^2-1}{n^2+1}\right)$$

Q21: A system consists of two types of gas molecules A and B having same number density $2 \times 10^{25} / m^3$. The diameter of A and B are 10 Å and 5 Å respectively. They suffer collision at room temperature. The ratio of average distance covered by the molecule A to that of B between two successive collision is ______ $\times 10^{-2}$

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25.00

Solution:

$$\lambda \propto \frac{1}{d^2}$$

$$\frac{\lambda_{10}}{\lambda_5} = \frac{1}{4} = 0.25$$

Q22: A light beam of wavelength 500 nm is incident on a metal having work function of 1.25 eV, placed in a magnetic field of intensity B. The electrons emitted perpendicular to the magnetic field B, with maximum kinetic energy are bent into circular arc of radius 30 cm. The value of B is ________$\times\ 10^{-7}\ T$.
Given $hc = 20 \times 10^{-26}\ J - m$, mass of electron $= 9 \times 10^{-31}\ kg$

Q22: A light beam of wavelength 500 nm is incident on a metal having work function of 1.25 eV, placed in a magnetic field of intensity B. The electrons emitted perpendicular to the magnetic field B, with maximum kinetic energy are bent into circular arc of radius 30 cm. The value of B is ________$\times 10^{-7}\ T$.
Given $hc = 20 \times 10^{-26}\ J-m$, mass of electron $= 9 \times 10^{-31}\ kg$

125.00

Solution:

$$\mathrm{R} = \frac{\sqrt{2m\mathrm{KE}}}{q\,\mathrm{B}} \text{ where KE} = \frac{hc}{\lambda} - 2 \times 10^{-19}$$

$$= \frac{\sqrt{2 \times 9 \times 10^{-31} \times 2 \times 10^{-19}}}{1.6 \times 10^{-19} \times \mathrm{B}}$$

$$\mathrm{B} = \frac{6 \times 10^{-25}}{0.3 \times 1.6 \times 10^{-19}} = 125 \times 10^{-7}\ \mathrm{T}$$

Q23: A message signal of frequency 20 kHz and peak voltage of 20 volt is used to modulate a carrier wave of frequency 1 MHz and peak voltage of 20 volt. The modulation index will be:

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1.00

Solution:

$$m=\frac{M}{A}=\frac{\text{modulation amplitude}}{\text{Carrier amplitude}}$$

$$=\frac{20}{20}=1$$

Q24: A 16Ω wire is bend to form a square loop. A 9V supply having internal resistance of 1Ω is connected across one of its sides. The potential drop across the diagonals of the square loop is __________ $\times\ 10^{-1}\ V$

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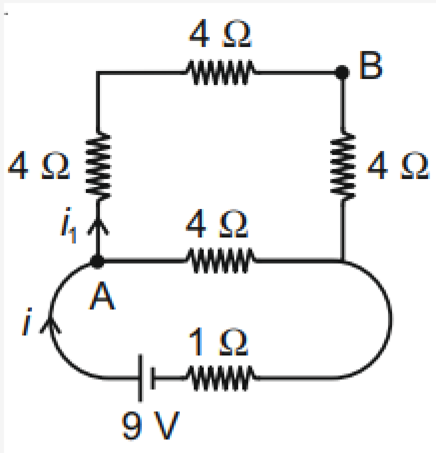
45.00

Solution:



$$i = \frac{9}{\frac{4\times12}{4+12}+1} = \frac{9}{4}\ \mathrm{A}$$

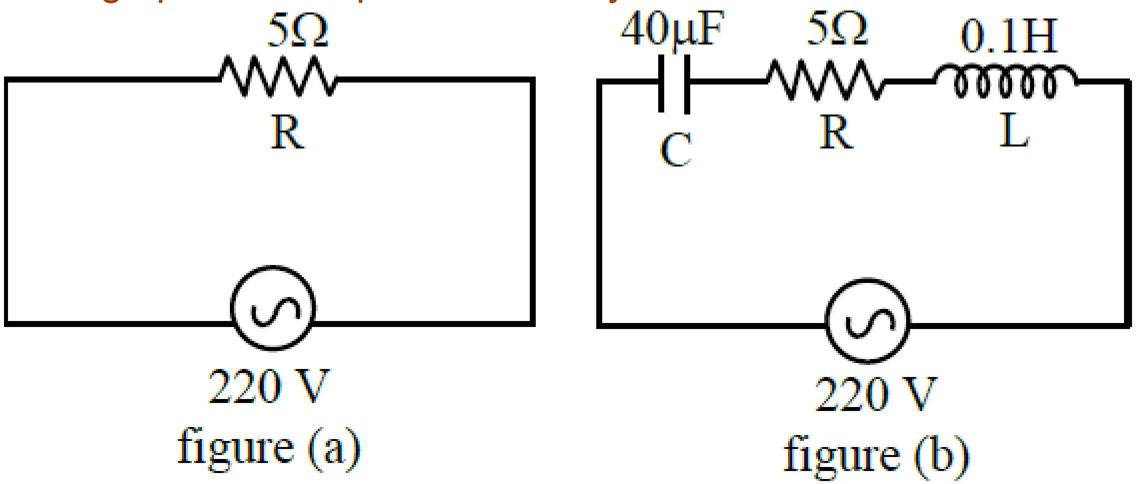
$$i_1 = \frac{9}{16}\ \mathrm{A}$$

$$\mathrm{V_A} - \mathrm{V_B} = i_1 \times \mathrm{R}$$

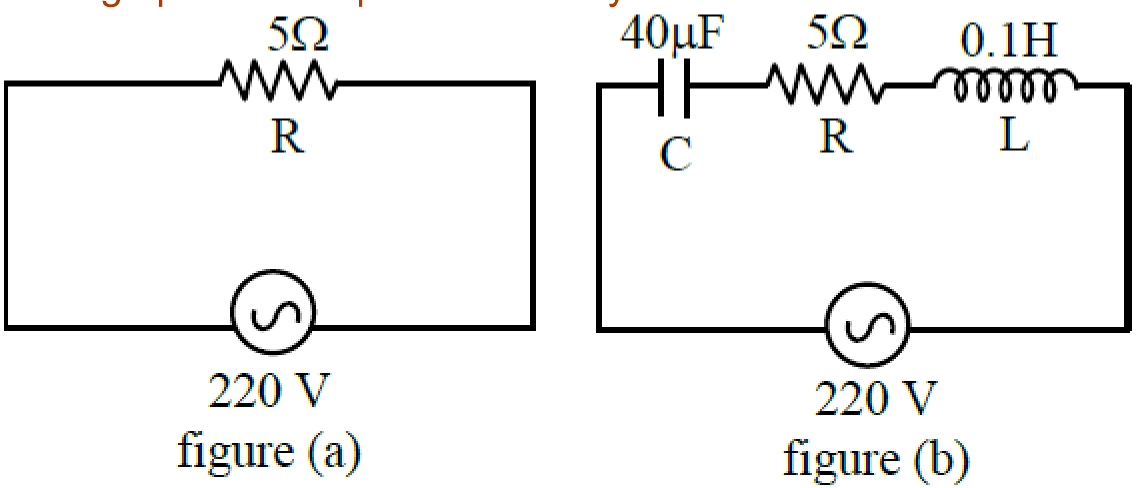
$$= \frac{9}{16}\cdot 8 = \frac{9}{2}\ \mathrm{V}$$

$$= 4.5\ \mathrm{V}$$

Q25: Two circuits are shown in the figure (a) & (b). At a frequency of ________ rad/s the average power dissipated in one cycle will be same in both the circuits.



Q25: Two circuits are shown in the figure (a) & (b). At a frequency of ________ rad/s the average power dissipated in one cycle will be same in both the circuits.



500.00

Solution:

$$\frac{V^2}{R} = \frac{V^2}{Z^2}R$$

$$\Rightarrow Z = R$$

$$\Rightarrow \omega = \frac{1}{\sqrt{LC}} = 500\mathrm{rad/s}$$

Q26: From the given data, the amount of energy required to break the nucleus of aluminium $^{27}_{13}A1$ is _____ $x \times 10^{-3}\ J$.

Mass of neutron = 1.00866 u

Mass of proton = 1.00726 u

Mass of Aluminium nucleus = 27.18846 u

(Assume 1 u corresponds to x J of energy)

(Round off to the nearest integer)

Q26: From the given data, the amount of energy required to break the nucleus of aluminium $^{27}_{13}A1$ is _____ $x \times 10^{-3}\ J$.

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Mass of proton = 1.00726 u

Mass of Aluminium nucleus = 27.18846 u

(Assume 1 u corresponds to x J of energy)

(Round off to the nearest integer)

27.00

Solution:

$$\text{B.E.} = [13(1.00726) + 14(1.00866) - 27.18846]x \text{ J}$$

$$[13.09438 + 14.12124 - 27.18846] \times \text{x J}$$

$$= (0.02716) \times \text{x J}$$

$$= 27.16 \times 10^{-3} \text{x J}$$

$$= 27 \times 10^{-3} \text{x J}$$

Q27: A force of $F = (5y + 20)\,\hat{j}$ N acts on a particle. The work done by this force when the particle is moved from y = 0 m to y = 10 m is _______ J.

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450.00

Solution:

$$W = \int \vec{F} \cdot d\vec{r}$$

$$= \int_0^{10} (5y + 20) dy$$

$$= 450 \text{ J}$$

Q28: A solid disc of radius 20 cm and mass 10 kg is rotating with an angular velocity of 600 rpm, about an axis normal to its circular plane and passing through its centre of mass. The retarding torque required to bring the disc at rest in 10 s is _____ $\pi \times 10^{-1}\ Nm$.

Q28: A solid disc of radius 20 cm and mass 10 kg is rotating with an angular velocity of 600 rpm, about an axis normal to its circular plane and passing through its centre of mass. The retarding torque required to bring the disc at rest in 10 s is _____ $\pi \times 10^{-1}\ Nm$.

4.00

Solution:

$$\Delta \mathrm{L} = \tau \Delta t$$

$$= \frac{\frac{1}{2} m \mathrm{R}^2 \omega}{10}$$

$$= \frac{\frac{1}{2} \times 10 \times 0.04 \times 600 \times 2\pi}{10 \times 60}$$

$$= \frac{24\pi}{60} = 0.4\pi$$

$$= 4 \times \pi \times 10^{-1}$$

Q29: In a semiconductor, the number density of intrinsic charge carriers at $27^{\circ} C \; is \; 1.5 \times 10^{16} / m^{3}$. If the semiconductor is doped with impurity atom, the hole density increases to $4.5 \times 10^{22} / m^{3}$. The electron density in the doped semiconductor is _______ $\times 10^{9} / m^{3}$.

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5.00

Solution:

$$n_e n_h = n_i^2$$

$$n_e = \frac{\left(1.5 \times 10^{16}\right)^2}{4.5 \times 10^{22}} = 5 \times 10^{9} / \mathrm{m}^{3}$$

Q30: The nuclear activity of a radioactive element becomes $\left(\frac{1}{8}\right)^{th}$ of its initial value in 30 years. The half-life of radioactive element is _________ years.

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10.00

Solution:

$$A = \frac{A_0}{2^n}$$

$$\Rightarrow n = 3$$

$$\Rightarrow T_{1/2} = 10 \text{ years}$$

Q31: In the following the correct bond order sequence is:

(A) $O_2^{2-} > O_2^{+} > O_2^{-} > O_2$

(B) $O_2^{+} > O_2^{-} > O_2^{2-} > O_2$

(C) $O_2^{+} > O_2 > O_2^{-} > O_2^{2-}$

(D) $O_2 > O_2^{-} > O_2^{2-} > O_2^{+}$

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(D) $O_2 > O_2^{-} > O_2^{2-} > O_2^{+}$

Solution:

According to Molecular orbital theory

$$\text{B.O.} = \frac{1}{2}[N_b - N_a]$$

$$O_2 = 2.0$$

$$O_2{}^{+} = 2.5$$

$$O_2{}^{-} = 1.5$$

$$O_2{}^{2-} = 1.0$$

Q32: A biodegradable polyamide can be made from:

(A) Glycine and isoprene

(B) Hexamethylene diamine and adipic acid

(C) Glycine and aminocaproic acid

(D) Styrene and caproic acid

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Solution:

Nylon 2-Nylon 6 (Polyamide copolymer) is biodegradable polymer.
Its monomer units are: Glycine + Aminocaproic acid
[H_2N-CH_2-COOH+NH_2(CH_2)-COOH]

Q33: Match **List I** with **List II -**

	List-I (Elements)		**List-II (Properties)**
(a)	Li	(i)	Poor water solubility of I^- salt
(b)	Na	(ii)	Most abundant element in cell fluid
(c)	K	(iii)	Bicarbonate salt used in fire extinguisher
(d)	Cs	(iv)	Carbonate salt decomposes easily on heating

Choose the correct answer from the options given below:

(A) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

(B) (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)

(C) (a)-(iv), (b)-(ii), (c)-(iii), (d)-(i)

(D) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

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(D) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

Solution:

(i) $Li_2CO_3 \xrightarrow{\Delta} Li_2O + CO_2$

(ii) $NaHCO_3$ is used in dry fire extinguishers.

(iii) Potassium has vital role in biological systems

(iv) CsI is least soluble due to smaller hydration enthalpy of its two ions.

Q34: Which one of the following metal complexes is most stable?

(A) $[Co(en)(NH_3)_4]Cl_2$

(B) $[Co(en)_3]Cl_2$

(C) $[Co(en)_2(NH_3)_2]Cl_2$

(D) $[Co(NH_3)_6]Cl_2$

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Solution:

For a complex having same central metal ion and coordination number, stability increases with the increase in the strength of the ligand. The order of strength is: en > NH_3. Thus, among the given complexes, the most stable would be $[Co(en)_3]Cl_2$. Also the compound has 3 chelate rings.

Q35: Which one of the following metal complexes is most stable?

	List-I		**List-II**
(a)	Concentration of Ag ore	(i)	Reverberatory furnace
(b)	Blast furnace	(ii)	Pig Iron
(c)	Blister copper	(iii)	Leaching with dilute NaCN
(d)	Froth floatation method	(iv)	Sulfide ores

Choose the correct answer from the options given below:

(A) (a)–(iii), (b)–(ii), (c)–(i), (d)–(iv)

(B) (a)–(iii), (b)–(iv), (c)–(i), (d)–(ii)

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(C) (a)–(iv), (b)–(i), (c)–(iii), (d)–(ii)

(D) (a)–(iv), (b)–(iii), (c)–(ii), (d)–(i)

Solution:

(a) Concentration of Ag ore is done by leaching with dilute NaCN –

$$4Ag + 8CN^- + 2H_2O + O_2 \rightarrow 4[Ag(CN)_2]^- + 4OH^-$$

$$2[Ag(CN)_2]^- + Zn \rightarrow [Zn(CN)_4]^{2-} + 2Ag$$

(b) The iron obtained from Blast furnace contains about 4% carbon and many impurities in smaller amount (e.g., S, P, Si, Mn). This is known as pig iron and cast into variety of shapes.

(c) The solidified copper obtained after process in reverberatory furnace has blistered appearance due to the evolution of SO_2 and so it is called blister copper.

(d) Froth flotation method is used for concentration of sulphide ores.

Q36: The ionic radii of F^{-} and O^{2-} respectively are 1.33 Å and 1.4 Å, while the covalent radius of N is 0.74 Å. The correct statement for the ionic radius of N^{3-} from the following is:

(A) It is smaller than F^{-} and N

(B) It is bigger than O^{2-} and F^{-}

(C) It is bigger than F^{-} and N, but smaller than of O^{2-}

(D) It is smaller than O^{2-} and F^{-}, but bigger than of N

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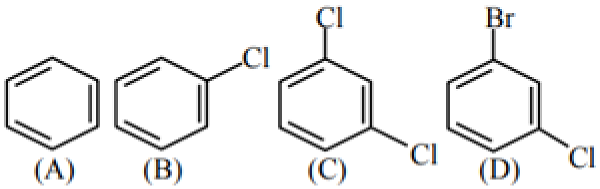
(C) It is bigger than F^- and N, but smaller than of O^{2-}

(D) It is smaller than O^{2-} and F^-, but bigger than of N

Solution:

F^-, O^{2-} and N^{3-} are isoelectronic. Size increases with increase in magnitude of negative charge. Thus, order will be $N^{3-} > O^{2-} > F^-$. Also, size of anions is bigger than the size of neutral atoms. Thus, $N^{3-} > N$.

Q37: The correct decreasing order of densities of the following compounds is :



(A) (D) > (C) > (B) > (A)

(B) (C) > (D) > (A) > (B)

(C) (C) > (B) > (A) > (D)

(D) (A) > (B) > (C) > (D)

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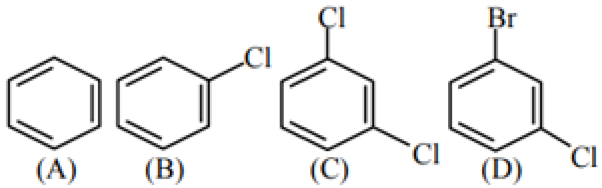


(A) (D) > (C) > (B) > (A)

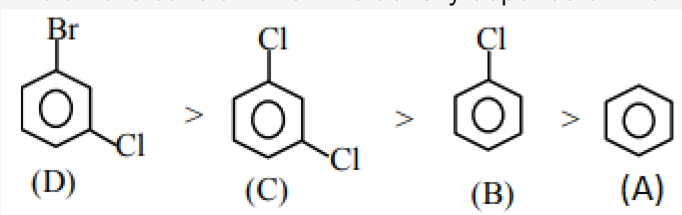
(B) (C) > (D) > (A) > (B)

(C) (C) > (B) > (A) > (D)

(D) (A) > (B) > (C) > (D)

Solution:

If volume is constant then the density depends on the molar mass of the compound.

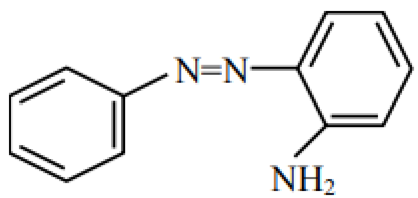


Q38: $$C_6H_5NO_2 \xrightarrow{Sn + HCl} \text{"A"} \xrightarrow[H^{\oplus}]{C_6H_5\overset{\oplus}{N}_2\overset{\ominus}{Cl}} P$$

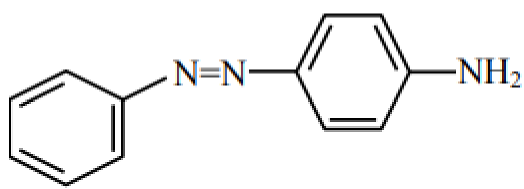
(Yellow coloured compound)

Consider the above reaction, the Product "P" is:

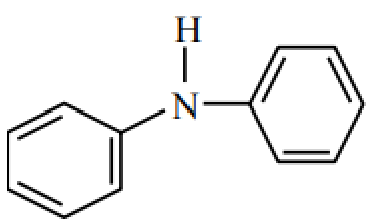
(A)



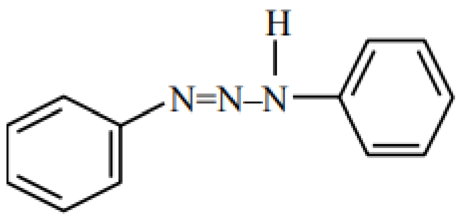
(B)



(C)



(D)



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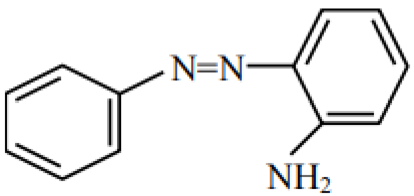
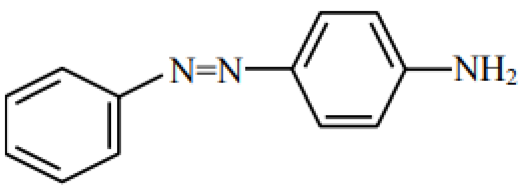
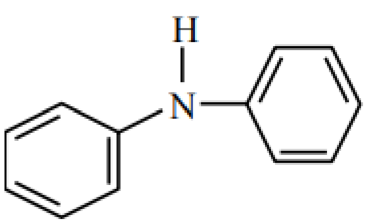


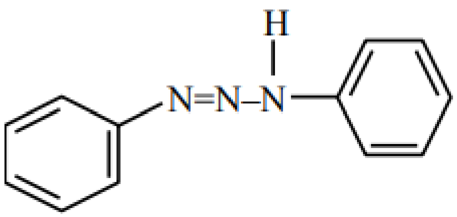
(B)



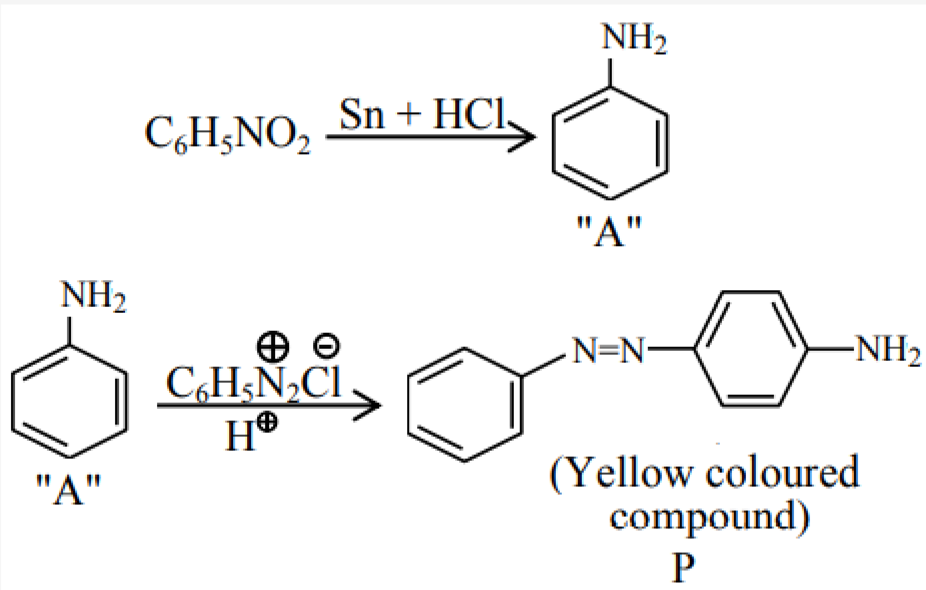
(C)



(D)


H
N=N–N

Solution:


NH2
C6H5NO2
Sn + HCl
"A"
NH2
C6H5N2Cl
H⊕
"A"
N=N
NH2
(Yellow coloured compound)
P

Q39: A reaction of benzonitrile with one equivalent CH_3MgBr followed by hydrolysis produces a yellow liquid "P". The compound "P" will give positive_____.

(A) Iodoform test

(B) Schiff's test

(C) Ninhydrin's test

(D) Tollen's test

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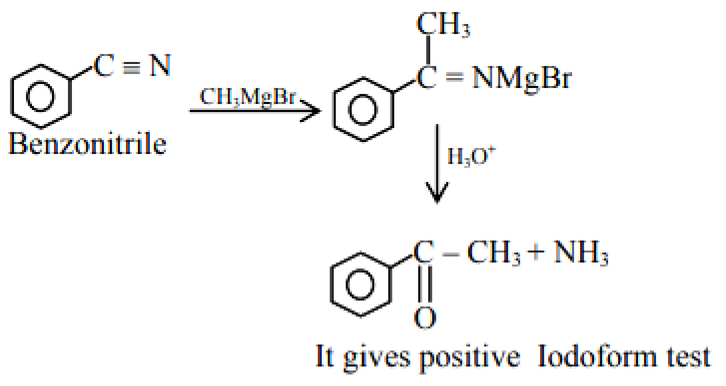

Q40: The spin only magnetic moments (in BM) for free Ti^{3+}, V^{2+} and Sc^{3+} ions respectively are (At. No. - Sc: 21, Ti: 22, V: 23)

(A) 3.87, 1.73, 0

(B) 1.73, 3.87, 0

(C) 1.73, 0, 3.87

(D) 0, 3.87, 1.73

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(A) 3.87, 1.73, 0

(B) 1.73, 3.87, 0

(C) 1.73, 0, 3.87

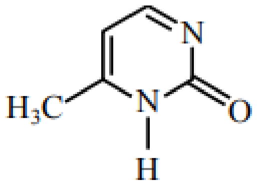
(D) 0, 3.87, 1.73

Solution:

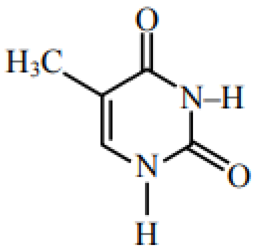
$$\mathrm{Ti}^{3+}\ \{\text{Unpaired electron} = 1\}$$
$$\mathrm{Sc}^{3+}\ \{\text{Unpaired electron} = 0\}$$
$$\mathrm{V}^{+2}\ \{\text{Unpaired electron} = 3\}$$
$$\mu = \sqrt{\mathrm{n}(\mathrm{n} + 2)}\,\mathrm{B.M}$$

Q41: Which one of the following is correct structure for cytosine?

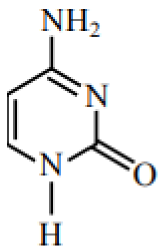
(A)



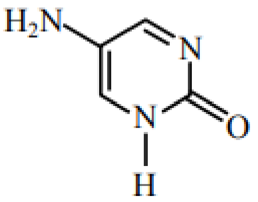
(B)



(C)

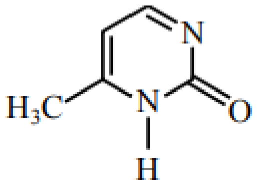


(D)

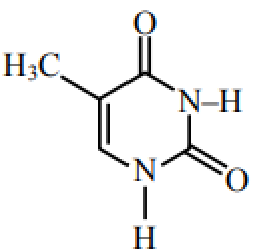


Q41: Which one of the following is correct structure for cytosine?

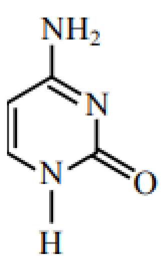
(A)



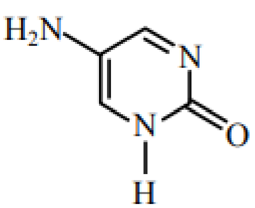
(B)



(C)

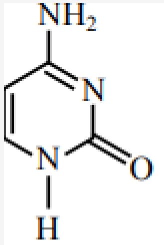


(D)



Solution:

Cytosine is –



Q42: Identify the species having one π-bond and maximum number of canonical forms from the following:

(A) SO_3

(B) O_2

(C) SO_2

(D) CO_3^{2-}

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(A) SO_3

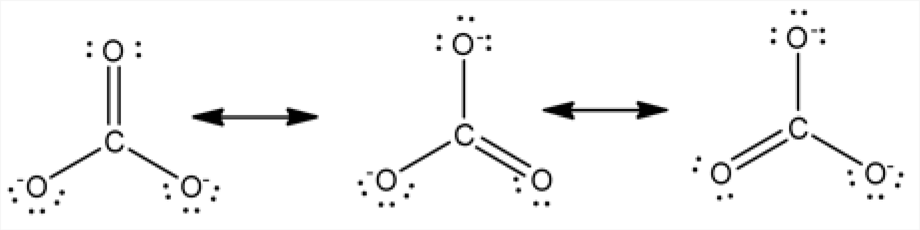
(B) O_2

(C) SO_2

(D) CO_3^{2-}

Solution:

(a) SO_3 – 3 pi bonds
(b) O_2 – 1 pi bond
(c) SO_2 – 2 pi bond
(d) CO_3^{2-} – 1 pi bond, 3 canonical forms.



Q43: Which one of the following metals forms interstitial hydride easily?

(A) Cr

(B) Fe

(C) Mn

(D) Co

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(A) Cr

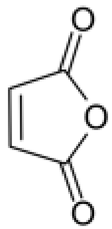
(B) Fe

(C) Mn

(D) Co

Solution:

These are formed by many d-block and f-block elements. However, the metals of group 7, 8 and 9 do not form hydride. Even from group 6, only chromium forms CrH.

Q44: 

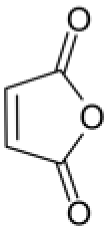
Maleic anhydride can be prepared by:

(A) Heating trans-but-2-enedioic acid

(B) Heating cis-but-2-enedioic acid

(C) Treating cis-but-2-enedioic acid with alcohol and acid

(D) Treating trans-but-2-enedioic acid with alcohol and acid

Q44: 

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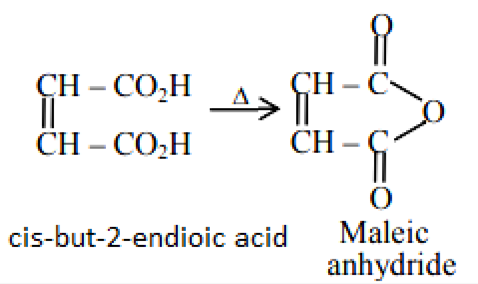
Solution:



Q45: Given below are two statements:

Statement I: Chlorofluorocarbons breakdown by radiation in the visible energy region and release chlorine gas in the atmosphere which then reacts with stratospheric ozone.

Statement II: Atmospheric ozone reacts with nitric oxide to give nitrogen and oxygen gases, which add to the atmosphere.

For the above statements choose the correct answer from the options given below:

(A) **Statement I** is incorrect, but statement II is true

(B) Both **statement I** and **II** are false

(C) **Statement I** is correct, but **statement II** is false

(D) Both **statement I** and **II** are correct

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(C) **Statement I** is correct, but **statement II** is false

(D) Both **statement I** and **II** are correct

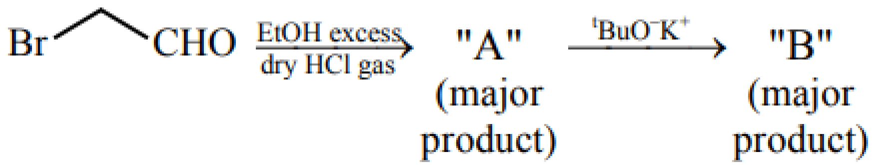
Solution:

Chlorofluorocarbons breakdown by powerful UV radiation and releases chlorine free radical which reacts with ozone and start chain reaction.

Atmosphere ozone reacts with nitric oxide to produce nitrogen dioxide and oxygen.

Hence both the statements are false statements.

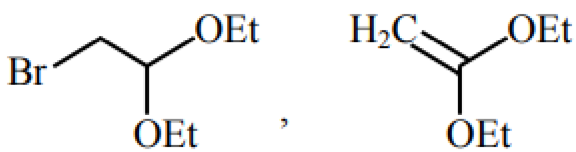
Q46:



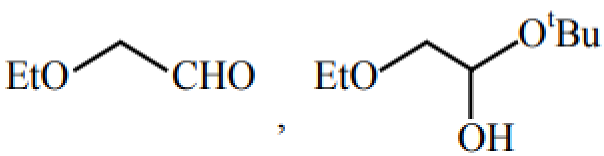
[where Et $\Rightarrow$ $-C_2H_5$ $^tBu \Rightarrow (CH_3)_3C-$]

Consider the above reaction sequence, Product "A" and Product "B" formed respectively are:

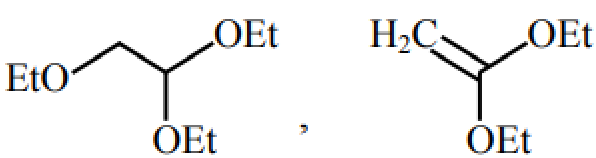
(A)



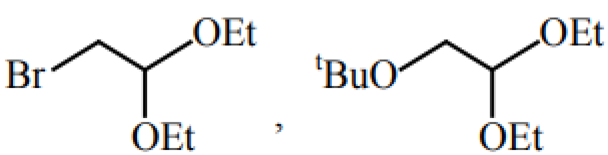
(B)

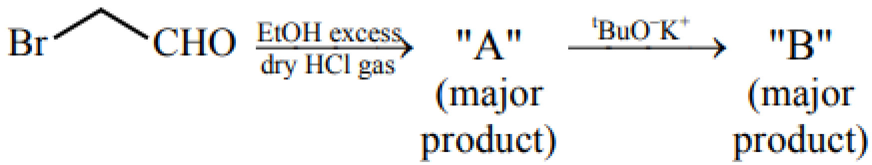


(C)



(D)



Q46: 

[where Et ⇒ $-C_2H_5$ ${}^tBu$ ⇒ $(CH_3)_3C-$]

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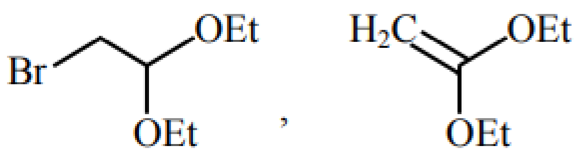
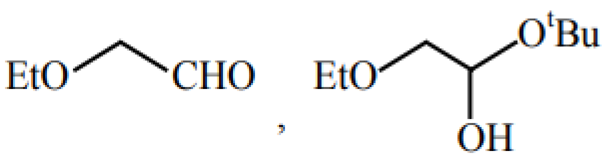
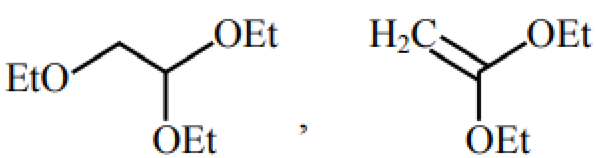
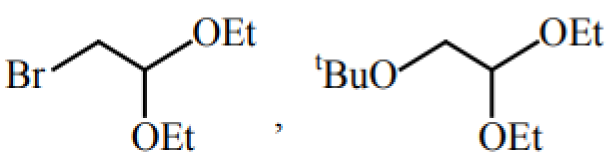
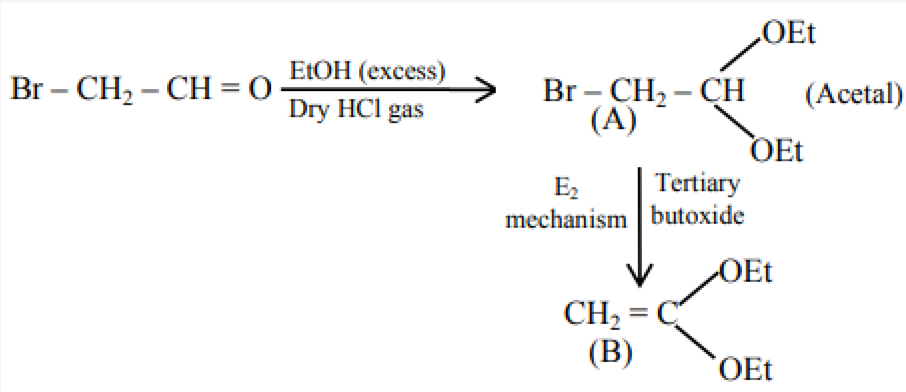
(A) 

(B) 

(C) 

(D) 

Solution:



Br – CH2 – CH = O
EtOH (excess)
Dry HCl gas
Br – CH2 – CH
OEt
OEt
(A)
(Acetal)
E2
mechanism
Tertiary
butoxide
CH2 = C
OEt
OEt
(B)

Q47: Match **List I** with **List II**:

List – I (Example of colloids)	**List – II (Classification)**
(a) Cheese	(i) dispersion of liquid in liquid
(b) Pumice stone	(ii) dispersion of liquid in gas
(c) Hair cream	(iii) dispersion of gas in solid
(d) Cloud	(iv) dispersion of liquid in solid

Choose the most appropriate answer from the options given below:

(A) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

(B) (a)-(iv), (b)-(i), (c)-(iii), (d)-(ii)

(C) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

(D) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)

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List – I (Example of colloids)	**List – II (Classification)**
(a) Cheese	(i) dispersion of liquid in liquid
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Choose the most appropriate answer from the options given below:

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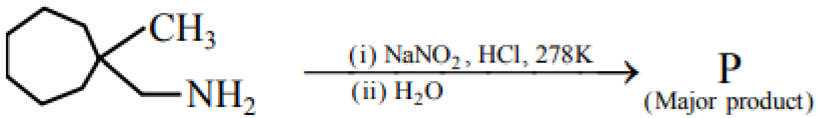
(C) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

(D) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)

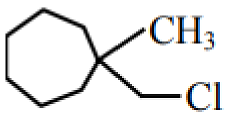
Solution:

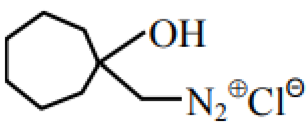
Dispersed Phase	**Dispersion medium**	Type of colloid	**Examples**
Solid	Solid	Solid sol	Some colored glasses and gemstones
Solid	Liquid	Sol	Paints, cell fluids
Solid	Gas	Aerosol	Smoke, dust
Liquid	Solid	Gel	Cheese, butter Jellies
Liquid	Liquid	Emulsion	Milk, hair cream
Liquid	Gas	Aerosol	Fog, mist, cloud, insecticide sprays
Gas	Solid	Solid sol	Pumice, stone, foam, rubber
Gas	Liquid	Foam	Froth, whipped, cream, soap lather

Q48: What is the major product "P" of the following reaction?



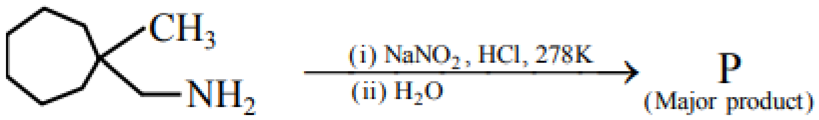
(A) 

(B) 

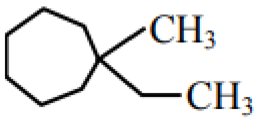
(C) 

(D) 

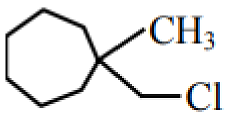
Q48: What is the major product "P" of the following reaction?



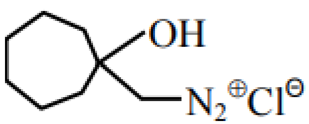
(A)



(B)

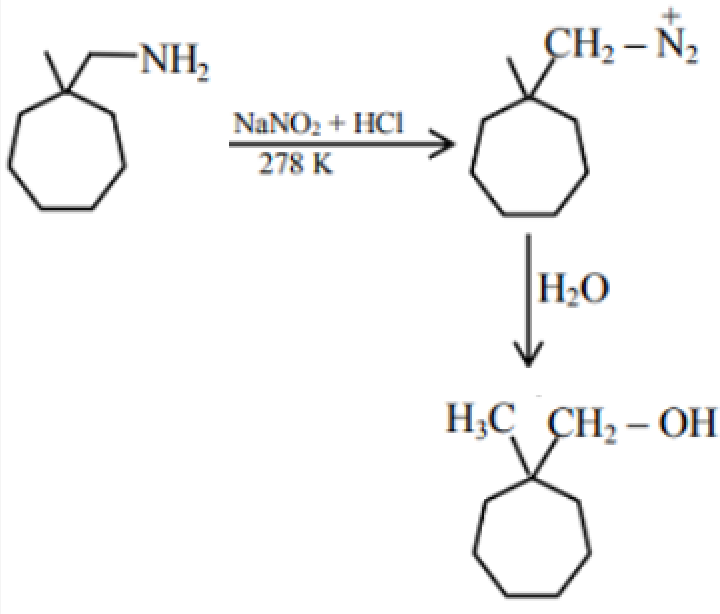


(C)



(D)

Solution:



(P)
Major
product

Q49: Identify the process in which change in the oxidation state is five:

(A) $Cr_2O_7^{2-} \rightarrow 2Cr^{3+}$

(B) $MnO_4^{-} \rightarrow Mn^{2+}$

(C) $CrO_4^{2-} \rightarrow Cr^{3+}$

(D) $C_2O_4^{2-} \rightarrow 2CO_2$

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(C) $CrO_4^{2-} \rightarrow Cr^{3+}$

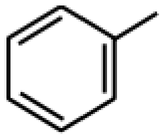
(D) $C_2O_4^{2-} \rightarrow 2CO_2$

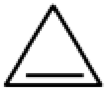
Solution:

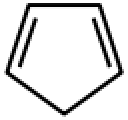
$Cr_2O_7^{2-} \rightarrow 2Cr^{3+}$ (Oxidation state changes from +6 to +3)
$MnO_4^{-} \rightarrow Mn^{2+}$ (Oxidation state changes from +7 to +2)
$CrO_4^{2-} \rightarrow Cr^{3+}$ (Oxidation state changes from +6 to +3)
$C_2O_4^{2-} \rightarrow 2CO_2$ (Oxidation state changes from +3 to +4)

Q50: Which among the following is the strongest acid?

(A) $CH_3CH_2CH_2CH_3$

(B) 

(C) 

(D) 

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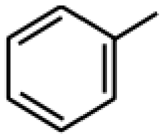
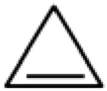
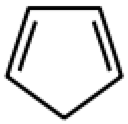
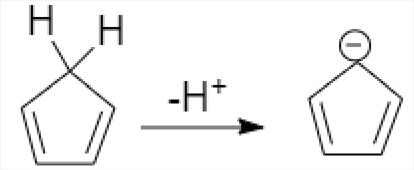
(B) 

(C) 

(D) 

Solution:

Conjugate base of compund (d) is aromatic hence it is most acidic compound.



Q51: A system does 200 J of work and at the same time absorbs 150 J of heat. The magnitude of the change in internal energy is ___________ J.

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50.00

Solution:

q = 150 J
w = –200 J

$$\Delta U = q + w = 150 - 200 = -50 \text{ J}$$

Q52: An accelerated electron has a speed of 5×10^6 m s^{-1} with an uncertainty of 0.02%. The uncertainty in finding its location while in motion is $x \times 10^{-9}$ m. The value of x is __. [Nearest integer]

[Use mass of electron = 9.1×10^{-31} kg, H = 6.63×10^{-34} Js, π = 3.14]

Q52: An accelerated electron has a speed of 5×10^{6} m s^{-1} with an uncertainty of 0.02%. The uncertainty in finding its location while in motion is $x \times 10^{-9}$ m. The value of x is ___. [Nearest integer]

[Use mass of electron = 9.1×10^{-31} kg, H = 6.63×10^{-34} Js, π = 3.14]

58.00

Solution:

$$\Delta v = \frac{0.02}{100} \times 5 \times 10^{6} = 10^{3} \text{ m s}^{-1}$$
$$\Rightarrow \Delta x . \Delta v \geqslant \frac{h}{4\pi m}$$
$$\Rightarrow x \times 10^{-9} \times 10^{3} \geqslant \frac{6.63 \times 10^{-34}}{4 \times 3.14 \times 9.1 \times 10^{-31}}$$
$$\Rightarrow x \geqslant \frac{6.63 \times 10^{-34}}{4 \times 3.14 \times 9.1 \times 10^{-31} \times 10^{-6}}$$
$$\Rightarrow x \geqslant 0.058 \times 10^{3}$$
$$\Rightarrow x \geqslant 58$$

Q53: Number of electrons present in 4f orbital of Ho^{3+} ion is ___________

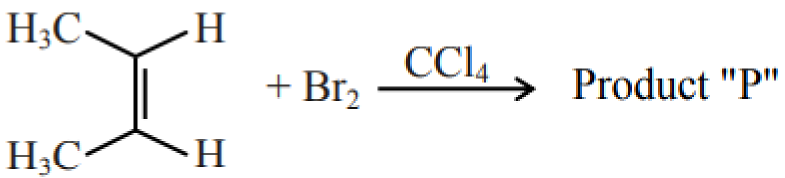
(Given Atomic No. of Ho = 67)

Q53: Number of electrons present in 4f orbital of Ho^{3+} ion is ___________
(Given Atomic No. of Ho = 67)

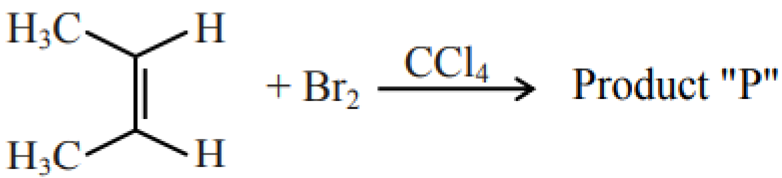
10.00

Solution:

$$Ho^{3+} = [Xe]\, 4f^{10}$$

Q54: 

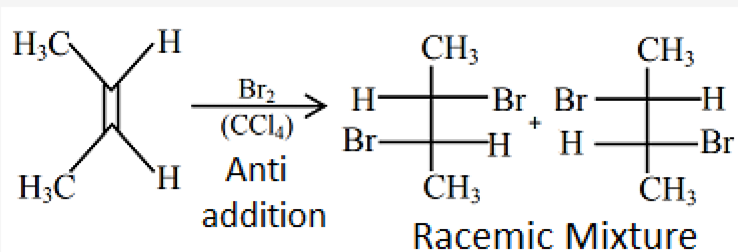
Consider the above chemical reaction. The total number of stereoisomers possible for Product 'P' is

Q54: 

Consider the above chemical reaction. The total number of stereoisomers possible for Product 'P' is

2.00

Solution:



Q55: For a chemical reaction A $\longrightarrow$ B, it was found that concentration of B is increased by 0.2 mol L^{-1} in 30 min. The average rate of the reaction is ___________ $\times 10^{-1}$ mol L^{-1} h^{-1}.

Q55: For a chemical reaction A $\longrightarrow$ B, it was found that concentration of B is increased by 0.2 mol L^{-1} in 30 min. The average rate of the reaction is ___________ $\times 10^{-1}$ mol L^{-1} h^{-1}.

4.00

Solution:

Rate = $\frac{[B]_2-[B]_1}{t_2-t_1} = \frac{0.2}{0.5} = 0.4 \text{ mol L}^{-1} \text{ h}^{-1}$

Q56: The number of significant figures in 0.00340 is ___.

Q56: The number of significant figures in 0.00340 is ___.

3.00

Solution:

There are certain rules for determining the number of significant figures. These are stated below:

(i) All non-zero digits are significant.

(ii) Zeros preceding to first non-zero digit are not significant.

(iii) Zeros at the end or right of a number are significant, provided they are on the right side of the decimal point.

Thus, the number of significant figures in 0.00340 is 3.

Q57: Assuming that $Ba(OH)_2$ is completely ionized in aqueous solution under the given conditions the concentration of H_3O^+ ions in 0.005 M aqueous solution of $Ba(OH)_2$ at 298 K is ___________ $\times 10^{-12}$ mol L^{-1} . (Nearest integer)

Q57: Assuming that $Ba(OH)_2$ is completely ionized in aqueous solution under the given conditions the concentration of H_3O^+ ions in 0.005 M aqueous solution of $Ba(OH)_2$ at 298 K is __________ $\times 10^{-12}$ mol L^{-1} . (Nearest integer)

1.00

Solution:

$[H_3O^+] = 0.005 \text{ M}$

Now, at 298 K, $[H_3O^+][OH^-] = 10^{-14}$

$\Rightarrow [OH^-] = \frac{10^{-14}}{0.005} = 2 \times 10^{-12}$

Now, $[Ba(OH)_2] = \frac{1}{2} \times [OH^-]$

$\Rightarrow [Ba(OH)_2] = \frac{1}{2} \times 2 \times 10^{-12} = 10^{-12}$

Q58: 0.8 g of an organic compound was analysed by Kjeldahl's method for the estimation of nitrogen. If the percentage of nitrogen in the compound was found to be 42%, then ___________ mL of 1 M H_2SO_4 would have been neutralized by the ammonia evolved during the analysis.

Q58: 0.8 g of an organic compound was analysed by Kjeldahl's method for the estimation of nitrogen. If the percentage of nitrogen in the compound was found to be 42%, then ___________ mL of 1 M H_2SO_4 would have been neutralized by the ammonia evolved during the analysis.

12.00

Solution:

$$\begin{aligned} \%\mathrm{N} &= \frac{1.4}{\mathrm{W}} \mathrm{N.V} \\ 42 &= \frac{1.4 \times (1 \times 2) \times \mathrm{V}}{0.8} \\ \mathrm{V} &= 12 \mathrm{~ml} \end{aligned}$$

Q59: When 3.00 g of a substance 'X' is dissolved in 100 g of CCl_4, it raises the boiling point by 0.60 K. The molar mass of the substance 'X' is _________ g mol^{-1}. (Nearest integer).
[Given - K_b for CCl_4 is 5.0 K kg mol^{-1}]

Q59: When 3.00 g of a substance 'X' is dissolved in 100 g of CCl_4, it raises the boiling point by 0.60 K. The molar mass of the substance 'X' is _________ g mol^{-1}. (Nearest integer).
[Given - K_b for CCl_4 is 5.0 K kg mol^{-1}]

250.00

Solution:

$$\Delta T_b = K_b m$$

$$= 5 \times \left(\frac{\text{Wt.} \times 1000}{\text{M.M} \times \text{Mass of solvent (g)}}\right)$$

$$= 5 \times \frac{3 \times 1000}{\text{M.M} \times 100}$$

$$0.6 = \frac{150}{\text{M.M}}$$

$$\text{M.M} = \frac{150}{0.6} = 250\,\text{g/mol}$$

Q60: An LPG cylinder contains gas at a pressure of 300 kPa at 27°C. The cylinder can withstand the pressure of 1.2×10^6 Pa. The room in which the cylinder is kept catches fire. The minimum temperature at which the bursting of cylinder will take place is ___________ °C. (Nearest integer)

Q60: An LPG cylinder contains gas at a pressure of 300 kPa at 27°C. The cylinder can withstand the pressure of 1.2×10^6 Pa. The room in which the cylinder is kept catches fire. The minimum temperature at which the bursting of cylinder will take place is __________ °C. (Nearest integer)

927.00

Solution:

Since the volume and amount of gas inside the cylinder will remain constant.

$$\frac{P}{T} = \text{constant}$$

$$\frac{P_1}{T_1} = \frac{P_2}{T_2} \Rightarrow \frac{300 \times 10^3}{300} = \frac{1.2 \times 10^6}{T}$$

$$T = \frac{1.2 \times 10^6}{10^3} = 1200 \text{ K} = 927^\circ \text{C}$$

Q61: Let $y = y(x)$ be the solution of the differential equation $xdy = \left(y + x^3 \cos x\right) dx$ with $y(\pi) = 0$, then $y\left(\frac{\pi}{2}\right)$ is equal to :

(A) $\frac{\pi^2}{2} - \frac{\pi}{4}$

(B) $\frac{\pi^2}{4} + \frac{\pi}{2}$

(C) $\frac{\pi^2}{4} - \frac{\pi}{4}$

(D) $\frac{\pi^2}{2} + \frac{\pi}{4}$

Q61: Let $y = y(x)$ be the solution of the differential equation $xdy = (y + x^3 \cos x)\, dx$ with $y(\pi) = 0$, then $y\left(\frac{\pi}{2}\right)$ is equal to :

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(B) $\frac{\pi^2}{4} + \frac{\pi}{2}$

(C) $\frac{\pi^2}{4} - \frac{\pi}{4}$

(D) $\frac{\pi^2}{2} + \frac{\pi}{4}$

Solution:

$xdy = (y + x^3 \cos x)\, dx$
$xdy = ydx + x^3 \cos x dx$
$\frac{xdy - ydx}{x^2} = \frac{x^3 \cos x dx}{x^2}$
$\int \frac{d}{dx}\left(\frac{y}{x}\right) dx = \int x \cos x dx$
$\Rightarrow \frac{y}{x} = x \sin x - \int 1. \sin x \; dx$
$\frac{y}{x} = x \sin x + \cos x + C$
When $x = \pi, y = 0$
$\Rightarrow 0 = -1 + C \Rightarrow C = 1$
So, $\frac{y}{x} = x \sin x + \cos x + 1$
$y = x^2 \sin x + x \cos x + x$
$y\left(\frac{\pi}{2}\right) = \frac{\pi^2}{4} + \frac{\pi}{2}$

Q62: The number of distinct real roots of $\begin{vmatrix} \sin x & \cos x & \cos x \\ \cos x & \sin x & \cos x \\ \cos x & \cos x & \sin x \end{vmatrix} = 0$ in the interval $-\frac{\pi}{4} \leqslant x \leqslant \frac{\pi}{4}$ is:

(A) 1

(B) 2

(C) 3

(D) 4

Q62: The number of distinct real roots of $\begin{vmatrix} \sin x & \cos x & \cos x \\ \cos x & \sin x & \cos x \\ \cos x & \cos x & \sin x \end{vmatrix} = 0$ in the interval $-\frac{\pi}{4} \leqslant x \leqslant \frac{\pi}{4}$ is:

(A) 1

(B) 2

(C) 3

(D) 4

Solution:

$$\begin{vmatrix} \sin x & \cos x & \cos x \\ \cos x & \sin x & \cos x \\ \cos x & \cos x & \sin x \end{vmatrix} = 0$$

Apply : $R_1 \rightarrow R_1 - R_2$ & $R_2 \rightarrow R_2 - R_3$

$$\begin{vmatrix} \sin x - \cos x & \cos x - \sin x & 0 \\ 0 & \sin x - \cos x & \cos x - \sin x \\ \cos x & \cos x & \sin x \end{vmatrix} = 0$$

$$(\sin x - \cos x)^2 \begin{vmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ \cos x & \cos x & \sin x \end{vmatrix} = 0$$

$(\sin x - \cos x)^2 (\sin x + 2\cos x) = 0$

$\sin x = \cos x$ or $\sin x = -2\cos x$

$\tan x = 1$ or $\tan x = -2$

As $-\frac{\pi}{4} \leqslant x \leqslant \frac{\pi}{4}, \therefore x = \frac{\pi}{4}$

Q63: Consider function $f : A \rightarrow B$ and $g : B \rightarrow C\ (A, B, C \subseteq R)$ such that $(gof)^{-1}$ exists, then:

(A) f and g both are one-one

(B) f is onto and g is one-one

(C) f is one-one and g is onto

(D) f and g both are onto

Q63: Consider function $f: A \to B$ and $g: B \to C\,(A, B, C \subseteq R)$ such that $(gof)^{-1}$ exists, then:

(A) f and g both are one-one

(B) f is onto and g is one-one

(C) f is one-one and g is onto

(D) f and g both are onto

Solution:

$\because \; (gof)^{-1}$ exists $\Rightarrow \; gof$ is bijective $\Rightarrow g\; f\,(x)$ must be bijective.
$\Rightarrow' \; f'$ must be one-one and ‘g’ must be onto.

Q64: The first of the two samples in a group has 100 items with mean 15 and standard deviation 3. If the whole group has 250 items with mean 15.6 and standard deviation $\sqrt{13.44}$, then the standard deviation of the second sample is:

(A) 5

(B) 8

(C) 4

(D) 6

Q64: The first of the two samples in a group has 100 items with mean 15 and standard deviation 3. If the whole group has 250 items with mean 15.6 and standard deviation $\sqrt{13.44}$, then the standard deviation of the second sample is:

(A) 5

(B) 8

(C) 4

(D) 6

Solution:

$n_1 = 100, \quad n = 250$

$\therefore \; n_2 = 250 - 100 \Rightarrow n_2 = 150$

$\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$

$15.6 = \frac{100(15)+(150)(\bar{x}_2)}{250}$

$\Rightarrow \; \boxed{\bar{x}_2 = 16}$

${\sigma_1}^2 = V_1(x) = 9, \sigma^2 = Var(x) = 13.44$

Let $V_2(x) = {\sigma_2}^2$

$$\boxed{\sigma^2 = \frac{n_1\sigma_1^2 + n_2\sigma_2^2}{n_1 + n_2} + \frac{n_1 n_2}{(n_1 + n_2)^2}(\bar{x}_1 - \bar{x}_2)^2}$$

$13.44 = \frac{100 \times 9 + 150 \times \sigma_2^2}{250} + \frac{100 \times 150}{(250)^2} \times 1$

$\Rightarrow \sigma_2^2 = 16 \Rightarrow \; \boxed{\sigma_2 = 4}$

Q65: The value of the integral $\int_{-1}^{1} \log\left(x + \sqrt{x^2+1}\right) dx$ is:

(A) 1

(B) 0

(C) –1

(D) 2

Q65: The value of the integral $\int\limits_{-1}^{1} \log\left(x+\sqrt{x^2+1}\right) dx$ is:

(A) 1

(B) 0

(C) –1

(D) 2

Solution:

Let $I = \int\limits_{-1}^{1} \log\left(x+\sqrt{x^2+1}\right) dx$

$\because$ log $\left(x+\sqrt{x^2+1}\right)$ is an odd function

$\therefore \int\limits_{-1}^{1} \ln\left(x+\sqrt{x^2+1}\right) dx = 0$

Q66: If $P = \begin{bmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{bmatrix}$, then P^{50} is:

(A) $\begin{bmatrix} 1 & 25 \\ 0 & 1 \end{bmatrix}$

(B) $\begin{bmatrix} 1 & 0 \\ 25 & 1 \end{bmatrix}$

(C) $\begin{bmatrix} 1 & 0 \\ 50 & 1 \end{bmatrix}$

(D) $\begin{bmatrix} 1 & 50 \\ 0 & 1 \end{bmatrix}$

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(B) $\begin{bmatrix} 1 & 0 \\ 25 & 1 \end{bmatrix}$

(C) $\begin{bmatrix} 1 & 0 \\ 50 & 1 \end{bmatrix}$

(D) $\begin{bmatrix} 1 & 50 \\ 0 & 1 \end{bmatrix}$

Solution:

$$P = \begin{bmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{bmatrix}$$

$$P^2 = \begin{bmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

$$P^3 = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \frac{3}{2} & 1 \end{bmatrix}$$

$$P^4 = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$$

$$\therefore P^{50} = \begin{bmatrix} 1 & 0 \\ 25 & 1 \end{bmatrix}$$

Q67: Let a, b and c be distinct positive numbers. If the vectors $a\hat{i} + a\hat{j} + c\hat{k}$, $\hat{i} + \hat{k}$ and $c\hat{i} + c\hat{j} + b\hat{k}$ are co-planar, then c is equal to:

(A) $\sqrt{ab}$

(B) $\frac{a+b}{2}$

(C) $\frac{1}{a} + \frac{1}{b}$

(D) $\frac{2}{\frac{1}{a} + \frac{1}{b}}$

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(A) $\sqrt{ab}$

(B) $\frac{a+b}{2}$

(C) $\frac{1}{a} + \frac{1}{b}$

(D) $\frac{2}{\frac{1}{a}+\frac{1}{b}}$

Solution:

Hence $\begin{vmatrix} a & a & c \\ 1 & 0 & 1 \\ c & c & b \end{vmatrix} = 0$

$\Rightarrow c^2 = ab \Rightarrow c = \sqrt{ab}$

Q68: Let X be a random variable such that the probability function of a distribution is given by $P(X=0)=\frac{1}{2}, P(X=j)=\frac{1}{3^j}(j=1,2,3,\ldots,\infty)$ .Then the mean of the distribution and P(X is positive and even) respectively are:

(A) $\frac{3}{4}$ and $\frac{1}{9}$

(B) $\frac{3}{4}$ and $\frac{1}{16}$

(C) $\frac{3}{8}$ and $\frac{1}{8}$

(D) $\frac{3}{4}$ and $\frac{1}{8}$

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(A) $\frac{3}{4}$ and $\frac{1}{9}$

(B) $\frac{3}{4}$ and $\frac{1}{16}$

(C) $\frac{3}{8}$ and $\frac{1}{8}$

(D) $\frac{3}{4}$ and $\frac{1}{8}$

Solution:

$$\text{Mean} = \sum X_i P_i = \sum_{r=0}^{\infty} r.\frac{1}{3^r} = \frac{3}{4}$$

$$P(X \text{ is even}) = \frac{1}{3^2} + \frac{1}{3^4} + \ldots\infty$$

$$= \frac{\frac{1}{9}}{1-\frac{1}{9}} = \frac{1}{8}$$

Q69: Consider the statement “The match will be played only if the weather is good and ground is not wet”. Select the correct negation from the following:

(A) The match will not be played and weather is not good and ground is wet.

(B) If the match will not be played, then either weather is not good or ground is wet.

(C) The match will not be played or weather is good and ground is not wet.

(D) The match will be played and weather is not good or ground is wet.

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(B) If the match will not be played, then either weather is not good or ground is wet.

(C) The match will not be played or weather is good and ground is not wet.

(D) The match will be played and weather is not good or ground is wet.

Solution:

p : weather is good
q : ground is not wet
$\sim (p \wedge q) \equiv \sim p \vee \sim q$
$\equiv$ weather is not good or ground is wet

Q70: If ${}^{n}P_{r} = {}^{n}P_{r+1}$ and ${}^{n}C_{r} = {}^{n}C_{r-1}$, then the value of r is equal to

(A) 3

(B) 1

(C) 4

(D) 2

Q70: If ${}^{n}P_{r} = {}^{n}P_{r+1}$ and ${}^{n}C_{r} = {}^{n}C_{r-1}$, then the value of r is equal to

(A) 3

(B) 1

(C) 4

(D) 2

Solution:

$${}^{n}p_{r} = {}^{n}p_{r+1} \Rightarrow \frac{n!}{(n-r)!} = \frac{n!}{(n-r-1)!}$$
$$\Rightarrow (n-r) = 1 \quad \ldots (1)$$
$${}^{n}C_{r} = {}^{n}C_{r-1}$$
$$\Rightarrow \frac{n!}{r!(n-r)!} = \frac{n!}{(r-1)!(n-r+1)!}$$
$$\Rightarrow \frac{1}{r(n-r)!} = \frac{1}{(n-r+1)(n-r)!}$$
$$\Rightarrow n-r+1 = r$$
$$\Rightarrow n+1 = 2r \quad \ldots (2)$$
From $(1) \Rightarrow 2r-1-r = 1 \Rightarrow r = 2$

Q71: If a tangent to the ellipse $x^2 + 4y^2 = 4$meets the tangents at the extremities of its major axis at B and C, then the circle with BC as diameter passes through the point:

(A) (–1, 1)

(B) (1, 1)

(C) $(\sqrt{3}, 0)$

(D) $(\sqrt{2}, 0)$

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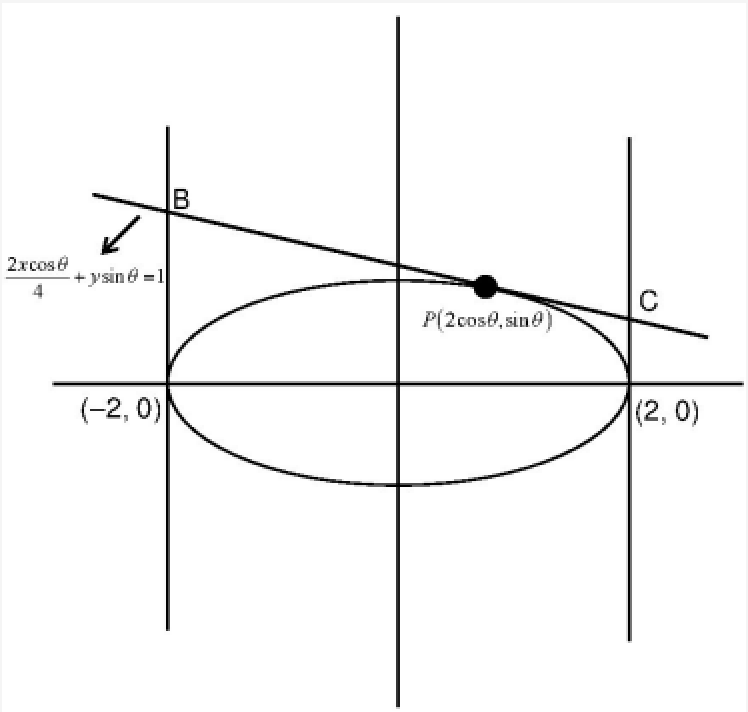
(A) (–1, 1)

(B) (1, 1)

(C) $\left(\sqrt{3}, 0\right)$

(D) $\left(\sqrt{2}, 0\right)$

Solution:



$\frac{x^2}{4}+\frac{y^2}{1}=1 \Rightarrow a=2$ & $b=1$

Let $P(2\cos\theta,\sin\theta)$

Equation of tangent at P is $(\cos\theta)x+2\sin\theta y=2$

$B\left(-2,\frac{1+\cos\theta}{\sin\theta}\right),\ C\left(2,\frac{1-\cos\theta}{\sin\theta}\right)$

$B\left(-2,\cot\frac{\theta}{2}\right)\ C\left(2,\tan\frac{\theta}{2}\right)$

Equation of circle is

$(x+2)(x-2)+\left(y-\cot\frac{\theta}{2}\right)\left(y-\tan\frac{\theta}{2}\right)=0$

$$x^2 - 4 + y^2 - \left(\tan\frac{\theta}{2} + \cot\frac{\theta}{2}\right) y + 1 = 0 \ \ldots (1)$$

So, $\left(\sqrt{3}, 0\right)$ satisfies equation (1)

Q72: The number of real solutions of the equation, $x^2 - |x| - 12 = 0$ is:

(A) 3

(B) 1

(C) 2

(D) 4

Q72: The number of real solutions of the equation, $x^2 - |x| - 12 = 0$ is:

(A) 3

(B) 1

(C) 2

(D) 4

Solution:

$$|x|^2 - |x| - 12 = 0$$
$$(|x| + 3)(|x| - 4) = 0$$
$$|x| = 4 \Rightarrow x = \pm 4 \quad (\because |x| \neq -3)$$

Q73: The sum of all those terms which are rational numbers in the expansion of $\left(2^{\frac{1}{3}}+3^{\frac{1}{4}}\right)^{12}$ is:

(A) 27

(B) 89

(C) 35

(D) 43

Q73: The sum of all those terms which are rational numbers in the expansion of $\left(2^{\frac{1}{3}}+3^{\frac{1}{4}}\right)^{12}$ is:

(A) 27

(B) 89

(C) 35

(D) 43

Solution:

$T_{r+1} = {}^{12}C_r\left(2^{\frac{1}{3}}\right)^r.\left(3^{\frac{1}{4}}\right)^{12-r}$

T_{r+1} will be rational number when $r=0,12$

$\therefore\ T_1+T_{13}=1\times 3^3+1\times 2^4\times 1=27+16=43$

Q74: If $|\vec{a}| = 2$, $\left|\vec{b}\right| = 5$ and $\left|\vec{a} \times \vec{b}\right| = 8$, then $\left|\vec{a}.\,\vec{b}\right|$ is equal to:

(A) 5

(B) 4

(C) 6

(D) 3

Q74: If $|\vec{a}| = 2, |\vec{b}| = 5$ and $|\vec{a} \times \vec{b}| = 8$, then $|\vec{a}.\vec{b}|$ is equal to:

(A) 5

(B) 4

(C) 6

(D) 3

Solution:

$|\vec{a}| = 2, |\vec{b}| = 5$
$|\vec{a} \times \vec{b}| = |\vec{a}| |\vec{b}| \sin\theta = 8$
$\sin\theta = \frac{4}{5}$
$\therefore \vec{a}.\vec{b} = |\vec{a}| |\vec{b}| \cos\theta = 10. \left(\pm\frac{3}{5}\right) = \pm 6$
$|\vec{a}.\vec{b}| = 6$

Q75: If the greatest value of the term independent of $'x'$ in the expansion of $\left(x\sin\alpha + a\frac{\cos\alpha}{x}\right)^{10}$ is $\frac{10!}{(5!)^2}$, then the value of $'a'$ is equal to:

(A) 2

(B) –1

(C) 1

(D) –2

Q75: If the greatest value of the term independent of $'x'$ in the expansion of $\left(x \sin \alpha + a \frac{\cos \alpha}{x}\right)^{10}$ is $\frac{10!}{(5!)^2}$, then the value of $'a'$ is equal to:

(A) 2

(B) –1

(C) 1

(D) –2

Solution:

$T_{r+1} = {}^{10}C_r (x \sin \alpha)^{10-r} \left(\frac{a \cos \alpha}{x}\right)^r$

T_{r+1} will be independent of x

When $10 - 2r = 0 \Rightarrow \boxed{r = 5}$

$T_6 = {}^{10}C_5 (x \sin \alpha)^5 \times \left(\frac{a \cos \alpha}{x}\right)^5$

$= {}^{10}C_5\ a^5 \times \frac{1}{2^5} (\sin 2\alpha)^5$

Will be greatest when $\sin 2\alpha = 1$

$\Rightarrow {}^{10}C_5 \frac{a^5}{2^5} = {}^{10}C_5 \Rightarrow \boxed{a = 2}$

Q76: The value of $\cot \frac{\pi}{24}$ is:

(A) $\sqrt{2}-\sqrt{3}-2+\sqrt{6}$

(B) $3\sqrt{2}-\sqrt{3}-\sqrt{6}$

(C) $\sqrt{2}-\sqrt{3}+2-\sqrt{6}$

(D) $\sqrt{2}+\sqrt{3}+2+\sqrt{6}$

Q76: The value of cot $\frac{\pi}{24}$ is:

(A) $\sqrt{2}-\sqrt{3}-2+\sqrt{6}$

(B) $3\sqrt{2}-\sqrt{3}-\sqrt{6}$

(C) $\sqrt{2}-\sqrt{3}+2-\sqrt{6}$

(D) $\sqrt{2}+\sqrt{3}+2+\sqrt{6}$

Solution:

$\cot\theta=\frac{1+\cos 2\theta}{\sin 2\theta}$ $\quad \left\{\begin{array}{l}\because 1+\cos 2\theta=2\cos^2\theta \\ \&\ \sin 2\theta=2\sin\theta\cos\theta\end{array}\right\}$

put, $\theta=\frac{\pi}{24}$

$\Rightarrow \cot\left(\frac{\pi}{24}\right)=\frac{1+\left(\frac{\sqrt{3}+1}{2\sqrt{2}}\right)}{\left(\frac{\sqrt{3}-1}{2\sqrt{2}}\right)}$ $\left\{\because\ \cos\frac{\pi}{12}=\frac{\sqrt{3}+1}{2\sqrt{2}}\&\ \sin\frac{\pi}{12}=\frac{\sqrt{3}-1}{2\sqrt{2}}\right\}$

$=\frac{(2\sqrt{2}+\sqrt{3}+1)}{(\sqrt{3}-1)}\times\frac{(\sqrt{3}+1)}{(\sqrt{3}+1)}$

$=\frac{2\sqrt{6}+2\sqrt{2}+3+\sqrt{3}+\sqrt{3}+1}{2}$

$=\sqrt{6}+\sqrt{2}+\sqrt{3}+2$

Q77: If $f(x) = \begin{cases} \int\limits_0^x (5 + |1 - t|)\, dt, & x > 2 \\ 5x + 1, & x \leqslant 2 \end{cases}$

(A) $f(x)$ is not differentiable at $x = 1$

(B) $f(x)$ is continuous but not differentiable at $x = 2$

(C) $f(x)$ is not continuous at $x = 2$

(D) $f(x)$ is everywhere differentiable

Q77: If $f(x) = \begin{cases} \int_0^x (5 + |1 - t|)\, dt, & x > 2 \\ 5x + 1, & x \leqslant 2 \end{cases}$

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(B) $f(x)$ is continuous but not differentiable at $x = 2$

(C) $f(x)$ is not continuous at $x = 2$

(D) $f(x)$ is everywhere differentiable

Solution:

For $x > 2$,

$$f(x) = \int_0^1 (5 + (1 - t))\, dt + \int_1^x (5 + (t - 1))\, dt$$

$$= 6 - \frac{1}{2} + \left(4t + \frac{t^2}{2}\right)\Big|_1^x$$

$$= \frac{11}{2} + 4x + \frac{x^2}{2} - 4 - \frac{1}{2}$$

$$= \frac{x^2}{2} + 4x + 1$$

$$f(2^+) = 2 + 8 + 1 = 11 = f(2^-) = f(2)$$

$\Rightarrow$ continuous at $x = 2$

Clearly differentiable at $x = 1$

$$Lf'(2) = 5$$

$$Rf'(2) = 6$$

$\Rightarrow$ not differentiable at $x = 2$

Q78: The lowest integer which is greater than $\left(1+\frac{1}{10^{100}}\right)^{10^{100}}$ is ______.

(A) 3

(B) 4

(C) 2

(D) 1

Q78: The lowest integer which is greater than $\left(1+\frac{1}{10^{100}}\right)^{10^{100}}$ is ______.

(A) 3

(B) 4

(C) 2

(D) 1

Solution:

Let $P=\left(1+\frac{1}{10^{100}}\right)^{10^{100}}$,

Let $x=10^{100}$

$\Rightarrow P=\left(1+\frac{1}{x}\right)^{x}$

$\Rightarrow P=1+(x)\left(\frac{1}{x}\right)+\frac{(x)(x-1)}{2!}\cdot\frac{1}{x^{2}}+\frac{(x)(x-1)(x-2)}{3!}\cdot\frac{1}{x^{3}}+\ldots$

(upto $10^{100}+1$ terms)

$\Rightarrow P=1+1+\left(\frac{1}{2!}-\frac{1}{2!x^{2}}\right)+\left(\frac{1}{3!}-\ldots\right)+..$ so on

$\Rightarrow P=2+\left(\text{Positive value less than }\frac{1}{2!}+\frac{1}{3!}+\frac{1}{4!}+\ldots\right)$

Also, $e=1+\frac{1}{1!}+\frac{1}{2!}+\frac{1}{3!}+\frac{1}{4!}+\ldots$

$\Rightarrow \frac{1}{2!}+\frac{1}{3!}+\frac{1}{4!}+\ldots=e-2$

$\Rightarrow P=2+$(Positive value less than $e-2)$

$\Rightarrow P\in(2,3)$

$\Rightarrow$ lowest integer which is greater than P is 3

Q79: If $[x]$ be the greatest integer less than or equal to x, then $\sum_{n=8}^{100}\left[\frac{(-1)^n n}{2}\right]$ is equal to:

(A) –2

(B) 4

(C) 2

(D) 0

Q79: If $[x]$ be the greatest integer less than or equal to x, then $\sum_{n=8}^{100}\left[\frac{(-1)^n n}{2}\right]$ is equal to:

(A) –2

(B) 4

(C) 2

(D) 0

Solution:

$$\sum_{n=8}^{100}\left[\frac{(-1)^n.n}{2}\right]=\left[\frac{8}{2}\right]+\left[\frac{-9}{2}\right]+\left[\frac{10}{2}\right]+\left[\frac{-11}{2}\right]+\ldots+\left[\frac{-99}{2}\right]+\left[\frac{100}{2}\right]$$
$$=4-5+5-6+6+\ldots-50+50=4$$

Q80: Let the equation of the pair of lines, $y = px$ and $y = qx$, can be written as $(y - px)(y - qx) = 0$. Then the equation of the pair of the angle bisectors of the lines $x^2 - 4xy - 5y^2 = 0$ is:

(A) $x^2 - 3xy - y^2 = 0$

(B) $x^2 + 3xy - y^2 = 0$

(C) $x^2 - 3xy + y^2 = 0$

(D) $x^2 + 4xy - y^2 = 0$

Q80: Let the equation of the pair of lines, $y = px$ and $y = qx$, can be written as $(y - px)(y - qx) = 0$. Then the equation of the pair of the angle bisectors of the lines $x^2 - 4xy - 5y^2 = 0$ is:

(A) $x^2 - 3xy - y^2 = 0$

(B) $x^2 + 3xy - y^2 = 0$

(C) $x^2 - 3xy + y^2 = 0$

(D) $x^2 + 4xy - y^2 = 0$

Solution:

$\left\{ \text{using formula for angle bisector of } ax^2 + 2hxy + by^2 = 0 \text{ as } \boxed{\dfrac{x^2 - y^2}{a - b} = \dfrac{xy}{h}} \right\}$

$\frac{x^2 - y^2}{1 - (-5)} = \frac{xy}{-2}$

$\frac{x^2 - y^2}{6} = \frac{xy}{-2}$

$\Rightarrow x^2 - y^2 = -3xy$

$\Rightarrow x^2 + 3xy - y^2 = 0$

Q81: If $a+b+c=1, ab+bc+ca=2$ and $abc=3$, then the value of $a^4+b^4+c^4$ is equal to ____.

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13.00

Solution:

$$a^2+b^2+c^2=(a+b+c)^2-2\sum ab=-3$$
$$(ab+bc+ca)^2=\Sigma(ab)^2+2abc\sum a$$
$$\Rightarrow \sum (ab)^2=-2$$
$$a^4+b^4+c^4=\left(a^2+b^2+c^2\right)^2-2\sum (ab)^2=9-2\left(-2\right)=13$$

Q82: Consider the function $f(x) = \begin{cases} \frac{P(x)}{\sin(x-2)}, & x \neq 2 \\ 7, & x = 2 \end{cases}$

where $P(x)$ is a polynomial such that $P''(x)$ is always a constant and $P(3) = 9$. If $f(x)$ is continuous at $x = 2$, then $P(5)$ is equal to

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39.00

Solution:

$$f(x) = \begin{cases} \frac{P(x)}{\sin(x-2)}, & x \neq 2 \\ 7, & x = 2 \end{cases}$$

$P''(x)$ is constant $\Rightarrow P(x)$ is a 2-degree polynomial

$f(x)$ is continuous at $x = 2$

$f(2^+) = f(2^-)$

$\lim_{x \to 2^+} \frac{P(x)}{\sin(x-2)} = 7$

So, $P(x) = (x-2)(ax+b)$

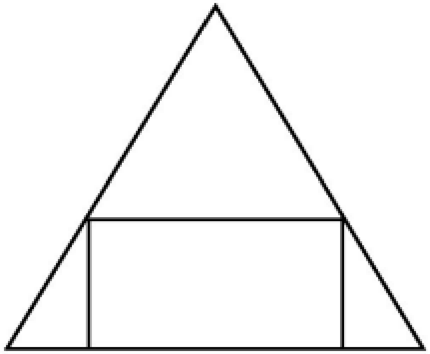
$\lim_{x \to 2^+} \frac{(x-2)(ax+b)}{\sin(x-2)} = 7 \Rightarrow \boxed{2a + b = 7}$

$P(3) = (3-2)(3a+b) = 9 \Rightarrow \boxed{3a + b = 9}$

$\boxed{a = 2, b = 3}$

$P(5) = (5-2)(2.5+3) = 3.13 = 39$

Q83: If a rectangle is inscribed in an equilateral triangle of side length $2\sqrt{2}$as shown in the figure, then the square of the largest area of such a rectangle is —



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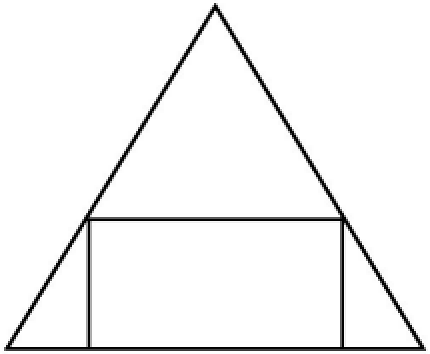
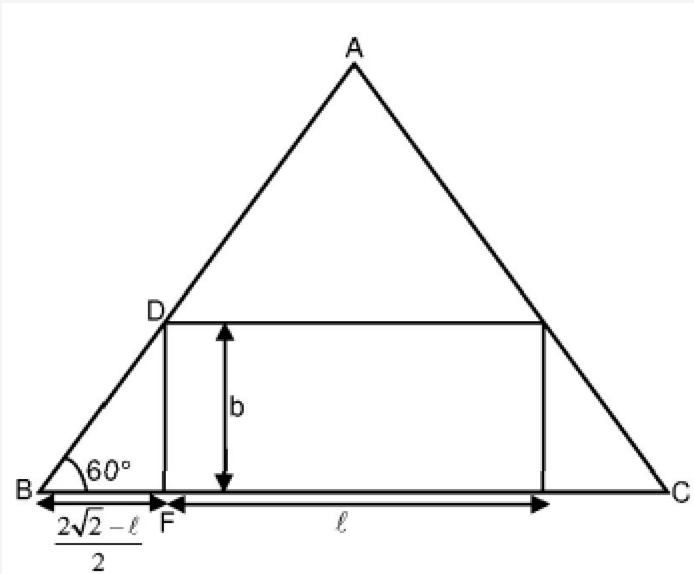


3.00

Solution:



In ΔDBF,

$$\tan 60^\circ = \frac{2b}{2\sqrt{2}-\ell} \Rightarrow \boxed{b = \frac{\sqrt{3}\left(2\sqrt{2}-\ell\right)}{2}}$$

$$A = \ell \times \frac{\sqrt{3}}{2}\left(2\sqrt{2}-\ell\right)$$

$$\frac{dA}{d\ell} = \frac{\sqrt{3}}{2}\left(2\sqrt{2}-2\ell\right)$$

For largest area of a rectangle, $\frac{dA}{dl} = 0$

$$\boxed{\ell = \sqrt{2}}$$

$$A = \ell \times b = \sqrt{2} \times \frac{\sqrt{3}}{2}\left(\sqrt{2}\right) = \sqrt{3}$$

$$\Rightarrow \boxed{A^2 = 3}$$

Q84: Let a curve $y=f\left(x\right)$ pass through the point $\left(2,\left(\log_e 2\right)^2\right)$ and have slope $\frac{2y}{x\log_e x}$ for all positive real value of x. Then the value of $f\left(e\right)$is equal to ___

Q84: Let a curve $y = f(x)$ pass through the point $\left(2, (\log_e 2)^2\right)$ and have slope $\frac{2y}{x\log_e x}$ for all positive real value of x. Then the value of $f(e)$is equal to ___

1.00

Solution:

$y' = \frac{2y}{x\log_e x} \Rightarrow \frac{dy}{dx} = \frac{2y}{x\log_e x}$
$\Rightarrow \frac{dy}{y} = \frac{2dx}{x\log_e x}$
$\Rightarrow \log_e |y| = 2\log_e |\log_e x| + C$
Put $x = 2, y = (\log_e 2)^2$
$\Rightarrow \log_e \left|(\log_e 2)^2\right| = \log_e \left|(\log_e 2)^2\right| + C$
$\Rightarrow C = 0$
$\Rightarrow y = (\log_e x)^2$
$\Rightarrow f(e) = 1$

Q85: A fair coin is tossed n-times such that the probability of getting at least one head is at least 0.9. Then the minimum value of n is _____ .

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4.00

Solution:

P (Head) $= \frac{1}{2}$

$1-\left(\frac{1}{2}\right)^{n} \geqslant 0.9$

$\Rightarrow \left(\frac{1}{2}\right)^{n} \leqslant \frac{1}{10}$

$\Rightarrow n_{\min} = 4$

Q86: The equation of a circle is $\operatorname{Re}\left(z^2\right) + 2(\operatorname{Im}(z))^2 + 2\operatorname{Re}(z) = 0$, where z = x + iy. A line which passes through the center of the given circle and the vertex of the parabola, $x^2 - 6x - y + 13 = 0$ has y-intercept equal to

Q86: The equation of a circle is $\mathrm{Re}\left(z^2\right)+2(\mathrm{Im}(z))^2+2\,\mathrm{Re}(z)=0$, where z = x + iy. A line which passes through the center of the given circle and the vertex of the parabola, $x^2-6x-y+13=0$ has y-intercept equal to

1.00

Solution:

Equation of circle is $\left(x^2-y^2\right)+2y^2+2x=0$
$x^2+y^2+2x=0$
Centre : $(-1,0)$
Parabola : $x^2-6x-y+13=0$
$(x-3)^2=y-4$
Vertex : $(3,4)$
So, equation of line is $y-0=\frac{4-0}{3+1}(x+1)$
$y=x+1$
y-intercept = 1

Q87: Let $n \in N$ and [x] denote the greatest integer less than or equal to x. If the sum of $(n+1)$terms of ${}^nC_0, 3.{}^nC_1, 5.{}^nC_2, 7.{}^nC_3$,.....is equal to $2^{100}.101$, then $2\left[\frac{n-1}{2}\right]$ is equal to________.

Q87: Let $n \in N$ and [x] denote the greatest integer less than or equal to x. If the sum of $(n+1)$terms of ${}^{n}C_0, 3.{}^{n}C_1, 5.{}^{n}C_2, 7.{}^{n}C_3,$.....is equal to $2^{100}.101$, then $2\left[\frac{n-1}{2}\right]$ is equal to________.

98.00

Solution:

$$S = 1.{}^{n}C_0 + 3.{}^{n}C_1 + 5.{}^{n}C_2 + \ldots + (2n+1)\ {}^{n}C_n$$
$$T_r = (2r+1)\ {}^{n}C_r$$
$$S = \sum T_r$$
$$S = \sum (2r+1)\ {}^{n}C_r = \sum 2r^{n}C_r + \sum {}^{n}C_r$$
$$S = 2\left(n.2^{n-1}\right) + 2^n = 2^n\,(n+1)$$
$$2^n\,(n+1) = 2^{100}.101 \Rightarrow n = 100$$

So, $2\left[\frac{n-1}{2}\right] = 2\left[\frac{99}{2}\right] = 98$

Q88: If the lines $\frac{x-k}{1} = \frac{y-2}{2} = \frac{z-3}{3}$ and $\frac{x+1}{3} = \frac{y+2}{2} = \frac{z+3}{1}$ are co-planar, then the value of k is ______________.

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1.00

Solution:

$$\begin{vmatrix} k+1 & 4 & 6 \\ 1 & 2 & 3 \\ 3 & 2 & 1 \end{vmatrix} = 0$$

{∵ Shortest distance between them is zero}

$$(k+1)[2-6]-4[1-9]+6[2-6]=0$$

$$k=1$$

Q89: If $\left(\vec{a}+3\vec{b}\right)$ is perpendicular to $\left(7\vec{a}-5\vec{b}\right)$ and $\left(\overrightarrow{a}-4\overrightarrow{b}\right)$ is perpendicular to $\left(7\vec{a}-2\vec{b}\right)$, then the angle between $\vec{a}$ and $\vec{b}$ (in degrees) is____

Q89: If $\left(\vec{a}+3\vec{b}\right)$ is perpendicular to $\left(7\vec{a}-5\vec{b}\right)$ and $\left(\overrightarrow{a}-4\overrightarrow{b}\right)$ is perpendicular to $\left(7\vec{a}-2\vec{b}\right)$, then the angle between $\vec{a}$ and $\vec{b}$ (in degrees) is___

60.00

Solution:

$$\left(\vec{a}+3\vec{b}\right)\perp\left(7\vec{a}-5\vec{b}\right)$$
$$\left(\vec{a}+3\vec{b}\right).\left(7\vec{a}-5\vec{b}\right)=0$$
$$7|\vec{a}|^2-15\left|\vec{b}\right|^2+16\vec{a}.\vec{b}=0\ \ldots(1)$$
$$\left(\vec{a}-4\vec{b}\right).\left(7\vec{a}-2\vec{b}\right)=0$$
$$7|\vec{a}|^2+8\left|\vec{b}\right|^2-30\vec{a}\cdot\vec{b}=0\quad\ldots(2)$$

From (1) and (2)

$$\boxed{|\vec{a}|=\left|\vec{b}\right|}$$
$$\therefore 7\,|\vec{a}|-15|\vec{a}|^2+16\vec{a}.\vec{b}=0$$
$$\Rightarrow \vec{a}.\vec{b}=\frac{|\vec{a}|^2}{2}$$
$$\therefore \cos\theta=\frac{\vec{a}.\vec{b}}{|\vec{a}|\left|\vec{b}\right|}=\frac{|\vec{a}|^2}{2|\vec{a}||\vec{a}|}$$
$$\therefore \cos\theta=\frac{1}{2}$$
$$\Rightarrow \theta=60^\circ$$

Q90: If the coefficients of x^7 and x^8 in the expansion of $\left(2+\frac{x}{3}\right)^n; n \in N$ are equal, then the value of n is ______.

Q90: If the coefficients of x^7 and x^8 in the expansion of $\left(2+\frac{x}{3}\right)^n; n \in N$ are equal, then the value of n is _____.

55.00

Solution:

$$T_{r+1} = \left({}^nC_r\right)(2)^{n-r}\left(\frac{x}{3}\right)^r = \left({}^nC_r\right)\left(\frac{2^{n-r}}{3^r}\right)x^r$$

According to the question, we have the following.

$$\left({}^nC_7\right)\left(\frac{2^{n-7}}{3^7}\right) = \left({}^nC_8\right)\left(\frac{2^{n-8}}{3^8}\right)$$

$$\Rightarrow \frac{{}^nC_7}{{}^nC_8} = \frac{2^{n-8}}{3^8} \times \frac{3^7}{2^{n-7}}$$

$$\Rightarrow \frac{n!}{7!\,(n-7)!} \times \frac{8!\,(n-8)!}{n!} = \frac{1}{6}$$

$$\Rightarrow \frac{8}{n-7} = \frac{1}{6}$$

$$\Rightarrow 48 = n - 7$$

$$\Rightarrow n = 55$$